CONVERGEO

Events Strategy & Ticketing Architecture

How the platform should think about, model, and discover events across Zambia

Companion Brief to the Strategic Master Plan

Builds on: 75 strategic decisions, dynamic-QR ticketing (Q50), all-events scope (Q49), the Services and Products briefs
Prepared: April 2026

1. How This Connects to What You've Already Decided

Your Strategic Master Plan locks in three decisions that shape everything in this brief. Q49 commits Convergeo to all event types — entertainment, conferences, workshops, private — from day one as part of the super-app behaviour. Q50 commits to in-app tickets with dynamic QR codes refreshing every 60 seconds. Q5 sets the commission model at 5–10% of ticket price. Day 49–53 of the roadmap allocates the build window.

That's the right foundation. What it doesn't yet specify is the model: what an Event actually is in the schema, how tickets relate to it, how non-ticketed and free events fit, how multi-day and recurring events work, what happens at the door, what happens when an event is cancelled, and which event categories should launch when. Without those decisions, the Day 49–53 work risks shipping a thin clone of every other ticketing platform — the same trap that the products brief warned about for catalogue.

> **Where this brief plugs in:** Section 4 of your Master Plan defines the Product/VendorListing schema and treats event tickets as a product class with a flag. This brief proposes adding a sibling Event entity (with TicketType and Ticket children) rather than overloading Product. The reason: events have time, capacity, and entry-control mechanics that don't compress cleanly into a product row. Same database, same APIs, parallel structure.

Why events are not products with dates

It's tempting to model an event ticket as a product with a date attribute and capacity = stock. It almost works. It breaks the moment any of the following happens, all of which are common in Zambia:

- The event has multiple ticket tiers (VIP, regular, student) with different prices, different perks, and different inventory pools — but a single shared event description, venue, and timing.
- The event runs for multiple days, and the customer either picks specific days or buys a pass that grants access to all days.
- The event is recurring — yoga every Tuesday at 6pm, monthly film screenings — and the platform should treat each instance as bookable while linking them under one event identity.
- The event is free but capacity-limited — kitchen parties, store openings, gallery exhibitions, religious gatherings — where RSVP and headcount tracking matter even though no money changes hands.
- The ticket is personalised — the buyer's name appears on the QR — for fraud prevention. Products do not behave this way.

Forcing all of this through the Product table creates either nightmare attribute JSON or a flood of phantom products. A small parallel Event/TicketType/Ticket schema solves it cleanly and lets the existing Cart, Order, Payment, and Escrow flows be reused with minimal change.

2. The Five Event Types

Every event on Convergeo collapses into one of five types. The type determines what fields the organiser must fill in, what the ticketing flow looks like, what entry control behaves like at the door, and what the refund and dispute policies are. This is the events equivalent of the six service archetypes and the five product classes.

Type	What it is	Examples and behaviour
Single-occurrence ticketed	One date, one start/end time, paid admission. Capacity-limited.	Concerts, comedy nights, sports fixtures, standalone workshops. The default. Dynamic QR per ticket. Escrow held until event date.
Multi-day ticketed	Same event identity spanning 2+ days. Customer may buy a day pass, multi-day pass, or full-festival pass.	Music festivals, agricultural fairs, multi-day expos, pop-up markets. Each ticket carries valid_dates[]; QR scans are date-aware.
Recurring	Same event template repeats on a schedule. Each instance is bookable.	Weekly yoga, monthly film series, regular comedy nights. One event template, many instances. Bookings attach to instance, not template.
Free RSVP / capacity-tracked	No payment, but capacity is finite and the organiser wants headcount and a guest list.	Store openings, religious gatherings, gallery openings, community meetings, kitchen parties. Free QR ticket; same entry-control mechanics; no escrow path.
Private / invitation-only	Not publicly listed. Discoverable only via direct link or invitation code.	Weddings, funerals (where ticketed seating is used), corporate events, private parties. Listed but unsearchable; access via signed URL or 6-digit code.

Three things follow:

- All five types share the same Event schema. The differences are captured in flags (is_recurring, is_private, is_free) and in child tables (TicketType, EventInstance for recurring). No type forks the codebase.
- Free RSVP events matter more than they look. Half the events on Lusaka's calendar — gallery openings, store launches, community gatherings — are free with RSVP. Excluding them means Convergeo only ever shows the paying half of the city's social calendar, and loses the discovery edge against Lusaka365 and Facebook events. They're cheap to support: same entry-control flow, no payment integration.
- Private events are a feature, not an edge case. Wedding ticketing for plate-paying receptions, funeral seating coordination, and corporate launches are real Zambian use cases. The mechanic — listed-but-unindexed, access via code — is the same code path as a normal event with one boolean flipped.

Schema implication (low cost, high payoff): Add an Event entity with type enum (single / multi_day / recurring / free_rsvp / private), an EventInstance child (date, start_time, end_time, capacity), a TicketType child (name, price, perks, inventory_per_instance), and a Ticket entity (the actual ticket purchased, with personalised holder_name, dynamic_qr_secret, status). One small subtree, all five types covered.

3. Ticket Types, Pricing, and Capacity

The ticketing layer is where most platforms over-engineer. Convergeo should ship six ticket-pricing modes from day one — even if only the simplest is exposed at launch — because retrofitting them later forces painful migrations.

3.1 The six ticket pricing modes

Mode	Description	When it applies
Fixed-price tier	One price, fixed inventory. "Regular admission, K150, 200 available."	The default. Almost all small events at launch.
Multi-tier	Multiple named tiers (VIP / Regular / Student) with separate prices and inventory pools.	Concerts, conferences, larger events. Each tier is a TicketType row under one Event.
Early-bird / time-staged	Same tier, price changes with time. "K100 until April 30, K150 after."	Use price_schedule on the TicketType. Auto-transitions on date.
Group / table	Sold as a bundle of N seats. "Table of 8 for K3,200."	Weddings, corporate dinners, private rooms at events. One sale, N tickets generated.
Free RSVP	Zero price, capacity tracked.	Type-4 events. Not a payment flow.
Pay-what-you-want	Customer chooses an amount within a min/max range.	Niche but useful for community events, fundraisers, religious gatherings. Phase 3.

3.2 Capacity is a tree, not a number

A naive model has Event.capacity = 500. The real world is messier:

- Total venue capacity (the hard ceiling, set by the organiser based on the venue).
- Per-tier capacity that sums to the total — VIP 50 + Regular 350 + Student 100 = 500.
- Per-day capacity for multi-day events — same tier may have different capacity each day.
- Held seats for organiser comps, sponsor allocations, and door sales (bookable through admin only).

Track capacity at the level where it's enforced: TicketType.inventory_total per instance, plus a venue-level safety check at checkout. Don't let tier-sums exceed venue capacity. This is one validation rule, easy to enforce at write time, that prevents a class of oversells that would destroy organiser trust on the first big event.

3.3 Fees: who pays and how it's framed

Q5 sets commission at 5–10% of ticket price. Two practical refinements:

- Display fees inclusive by default. The customer sees "K150" and at checkout sees "K150 ticket + K9 platform fee = K159" with a clear breakdown. Hidden fees that surface only at checkout are the single biggest cause of cart abandonment in ticketing — and Zambian buyers are particularly sensitive after the WhatsApp "send me K150 and I'll bring the ticket" alternative.
- Let organisers choose absorb-fee or pass-through. Big organisers will prefer absorbing the fee to keep round-number prices ("K150" not "K159"). Small organisers running thin margins will pass through. The platform takes the same cut either way; only the tax-incidence framing changes for the buyer.
- Set the floor commission at 5% (Q5 minimum) for the first 12 months. This matches the rate established ticketing platforms charge in the region and removes "too expensive" as a reason for organisers to stay on Facebook events.

3.4 Payment timing and escrow

Events are unique in your platform: there's a long gap between purchase and "delivery" (the event itself), sometimes weeks. This creates organiser cash-flow problems if escrow logic from products is applied naively.

- For events under 14 days away: hold full settlement until 24 hours after the event ends. Standard escrow.
- For events 14+ days away: release 50% of settlement to the organiser at T-7 days, the remaining 50% at T+1 day after the event. This funds the organiser's last-week production costs (staging, sound, security, catering deposits) without exposing the platform to a no-show event scam.
- For free RSVP events: no escrow path, but the platform still issues tickets and tracks attendance. Useful data for the city-guide AI assistant later (Q52).

4. Event Discovery: Time-First, Not Distance-First

The services brief argued for vector + geo search with proximity re-ranking. Events invert that. Distance still matters, but time is the dominant axis: a perfect event 5km away that happened yesterday is useless; an event 30km away tomorrow night is interesting. The discovery layer needs different defaults.

4.1 Three primary lenses

The events tab should expose three top-level lenses, all bookmarkable and shareable:

- When — Tonight, This Weekend, Next Week, Next Month, On a Date. Time-first browsing. This is what people actually want to know: "what's on this Saturday?"
- What — Music, Comedy, Workshops, Conferences, Sports, Family, Religious, Markets, Exhibitions. The category tree, used as filters layered on top of When.
- Where — by city (Lusaka, Ndola, Kitwe, Livingstone, Kabwe), by neighbourhood for dense cities, plus a "near me" radius option.

4.2 Search ranking signals

Hybrid retrieval (Meilisearch + pgvector, same as products and services) with an event-tuned re-ranking score:

- Time decay — events further in the future score lower than events imminent, modelling actual user intent. Events past their end-time drop off the index entirely.
- Sell-through pressure — events that are 70%+ sold get a small "selling fast" boost and a visible badge. This drives FOMO conversion and is the kind of signal that becomes invaluable once the platform has a few hundred events.
- Distance — same Haversine treatment as services, but weighted lower. People will travel further for the right event than they will for a haircut.
- Category match and follow signals — if a user has bought music tickets before, music ranks higher for them. Cheap personalisation, no ML required.
- Verification — verified organisers (Tier 2/3 KYC) outrank Tier 1 organisers. This is doubly important for events because the cold-start fraud risk is high.

4.3 Browse-by-time defaults

The homepage events module should default to "Tonight + This Weekend" rather than "Trending" or "All". This sounds minor; it isn't. It's the single decision that makes the events tab feel alive on first visit, because there are always events tonight or this weekend somewhere in Lusaka. "All upcoming events" feels like a directory; "Tonight" feels like a scene.

4.4 The calendar view (Phase 2)

After Phase 1 has 50+ active events, a calendar view becomes valuable: a month grid where each day shows event-density dots, expandable into the day's events. This is how power users (event-goers in Lusaka, tour planners, lifestyle bloggers) use a city's event ecosystem. It also feeds directly into the city-guide AI trip planner from Q52 — "show me everything happening in Livingstone the weekend I'm there".

4.5 The city-guide integration

Q52 commits to city guides for Lusaka, Livingstone, Ndola with AI enhancement later. Events should plug into city guides as a first-class section, not an afterthought. A Livingstone city guide page that shows tour operators (services), craft markets (products), and tonight's events at the local lodge — all bookable in one flow — is the actual super-app behaviour your master plan promised. None of the existing event platforms in Zambia bundle commerce, services, and event tickets in one transaction layer; this is where Convergeo's positioning crystallises.

5. Tickets, Dynamic QR, and Entry Control

Q50 commits to dynamic QR codes that refresh every 60 seconds. That's the right call — and it earns its complexity at the door, where 90% of ticketing fraud actually happens. This section spells out the mechanics so the Day 49–53 build doesn't ship a static-QR system that everyone refers to as "dynamic".

5.1 What dynamic QR actually means

Each issued Ticket carries a server-side secret (a long random string, never shown). The QR code displayed in the customer's app is generated client-side every 60 seconds by HMAC-signing (current_time_window || ticket_id) with that secret, encoded into a short URL or token. The scanner at the door hits a verification endpoint with the scanned token; the server checks the HMAC against the current and previous time-window using the same secret and returns valid / used / invalid / expired.

Three properties this gives you for free:

- A screenshot of the QR is useless after the next 60-second window. Scalpers cannot mass-share screenshots in WhatsApp groups.
- A re-scan of the same ticket fails after the first valid scan. Tickets cannot be used twice — even by the legitimate buyer who tries.
- The scanner works offline for short periods. Cache the day's secrets to the scanning device; sync "used" status when the network returns. Critical for venues in the Copperbelt where connectivity at the door is unreliable.

5.2 The PIN backup (Q33)

Q33 calls for QR primary + PIN backup. The PIN is a 6-digit code unique per ticket, displayed alongside the QR. If the camera fails (cracked phone screen, lighting at the door, dust on lens), the door staff types the PIN into the scanner app and the same verification logic runs. This is the kind of detail that quietly differentiates the platform on the very first event when something goes wrong, which it always does.

5.3 Personalisation and transfer

Each ticket carries a holder_name, captured at purchase. By default, this is the buyer; for group purchases, the buyer is asked to enter each holder's name.

- Tickets are transferable up to 6 hours before the event by the original buyer. Transfer revokes the old ticket and issues a new one to the recipient via SMS / WhatsApp. The new holder has their own QR + PIN.
- Transfer disabled within 6 hours of event start. Reduces same-day scalping; the legitimate "my friend got sick, take my ticket" case is still served by the door staff scanning the original buyer's ID against holder_name.

- Resale is blocked at platform level. Q-style scalping markets are how trust evaporates. If a buyer wants out, they can request refund per the policy below; they cannot list the ticket for sale.

5.4 Entry-control flow at the door

The vendor / organiser uses the same Convergeo app, switched into "event scanner" mode. Two-handed flow:

- Door staff opens the event in scanner mode → camera activates → scans QR → screen shows green check + holder name + tier ("Sarah Mwansa — VIP") in 1 second. Or red X with reason ("already used at 19:42", "expired", "wrong event").
- Organiser dashboard shows a live attendance counter: tickets sold, scanned, on-premises (sold minus scanned), capacity. Useful for security limits and air-quality decisions in indoor venues.
- Manual override: organiser-role users (not just staff) can mark a ticket as scanned in the dashboard if hardware fails entirely. Audit-logged.

5.5 Refunds, cancellations, and rescheduling

This is where event policies need to be explicit, not buried in T&Cs. A clear, customer-facing matrix builds far more trust than tickets refreshing every 60 seconds:

Scenario	Buyer entitled to	Mechanic
Buyer cancels >7 days before	Full refund minus 5% admin fee	Self-service via order page; auto-refund from escrow.
Buyer cancels 1–7 days before	50% refund or transfer	Self-service; organiser sees the cancellation.
Buyer cancels <24 hours before	No refund; transfer still allowed	Reduces last-minute fraud; transfer preserves goodwill.
Organiser cancels event	100% refund + automatic notification	Platform-initiated. Escrow returned in full. Organiser pays platform a small admin fee.
Organiser reschedules event	Tickets auto-honour the new date; opt-out refund available	Notification + 7-day window to request refund. Default keeps the ticket.
Force majeure (weather, etc.)	Negotiated case-by-case; platform mediates	Escrow held. Dispute flow (Q62).

6. Organiser Onboarding and Tooling

Event organisers are a different vendor archetype from product sellers and service providers. They're typically one-off or seasonal users — a person running an annual festival, a venue that hosts six events a year, a workshop facilitator running quarterly sessions. Onboarding has to be lighter than vendor onboarding for products, but trust requirements are arguably higher because event fraud is more visible and more reputationally damaging.

6.1 Two organiser tiers, mapped to your tiered KYC

The Q35/Q36 tiered KYC model maps cleanly onto organisers:

- Tier 1 — Personal organiser. Identity verified via NRC photo. Can run free RSVP events (Type 4) immediately and ticketed events up to a capped value (e.g. K20,000 GMV per event) until trust is built. After 3 successful events, cap lifts.
- Tier 2 — Business organiser. PACRA verified. Full ticketed events, no caps. Can run multi-day and recurring events. Earns the verified badge that customers see on every event card.

This compromise lets a one-person workshop facilitator launch instantly while reserving headline-act ticketing capacity for vendors with skin in the game.

6.2 The event creation flow

Three-screen flow, designed to take under 5 minutes for a simple event:

- Screen 1 — Basics: title, category, type (single / multi-day / recurring / free / private), one-line description, cover image, venue (with GPS pin), date(s) and time(s).
- Screen 2 — Tickets: add ticket types (name, price, capacity, perks). Free RSVP shows zero-price as the default. Multi-tier reveals an "add another tier" button.
- Screen 3 — Promotion & policy: longer description with rich text, social links, refund policy (defaults to platform standard), promotional banner, organiser bio. Organiser can publish or save as draft.

Save-as-draft matters more for events than for products. Organisers iterate on event copy, get feedback from co-organisers, and publish only when ready. The platform's bounce rate on event creation will be terrible if every form requires immediate publication.

6.3 Co-organisers and team access

Real events have multiple people involved: the organiser, the venue manager, door staff, sometimes a marketing partner. Each needs different access. A simple role model handles this without becoming an HR system:

- Owner — full edit, financial, and refund access. The person who created the event.
- Manager — can edit event details, view sales, but not pull funds or issue refunds.

- Door — scanner mode only. Cannot see financials, cannot edit anything.

Roles are added by phone number; the invited person gets an SMS with a one-tap link that grants them access scoped to that event only. This is one feature that quietly punches above its weight in differentiating from amateur ticketing.

6.4 Promotion tools the platform should provide

Don't expect organisers to be marketers. The platform should ship the basics:

- Auto-generated shareable link with rich Open Graph preview (title, image, date, price). One tap from the event page; works in WhatsApp, Facebook, X, Instagram bio.
- Promo codes — flat or percentage discount, capped quantity, expiry date. Used for influencer partnerships (Q28), sponsor giveaways, and "members get 20% off" mechanics.
- Affiliate links — a unique URL that tracks sales attributed to a specific influencer or partner, with optional automatic revenue share. This is the same primitive that makes Q28's TikTok / Instagram strategy actually measurable.
- Email/SMS to past attendees — once an organiser has run 2+ events, they can message past buyers about the next one. Opt-out respected. Quiet retention engine.

6.5 Analytics organisers actually use

Don't recreate Google Analytics. Show four numbers prominently on the organiser dashboard:

- Tickets sold vs capacity, by tier.
- Sales velocity — sales per day, with a trend arrow.
- Where buyers came from — direct, search, shared link, promo code (this is the only piece organisers will obsess over).
- Predicted final attendance — a simple "based on current pace, you'll sell ~340 of 500" projection. Wrong sometimes, useful always, and a reason to keep checking the dashboard.

7. Fraud and Trust Specific to Events

Event fraud has two distinct vectors that don't apply to products or services. Both need explicit defences.

7.1 Buyer-side fraud (counterfeit and duplicate tickets)

This is what dynamic QR solves. But two layered defences make it bulletproof:

- First-scan-wins, with logging. The first valid scan marks the ticket used. Subsequent scans return red with the original scan timestamp visible. No ambiguity at the door.
- ID match for high-value tickets (>K500). The QR scanner displays the holder name; door staff can be instructed to spot-check ID. Optional, organiser-toggleable. Disabled by default to avoid friction at small events.

7.2 Organiser-side fraud (no-show events)

Far more dangerous to platform reputation: an organiser collects ticket money for an event that never happens. Three mechanics in combination handle this:

- Tier 1 caps. New organisers cannot sell more than K20,000 of tickets for a single event until they've successfully run three. This caps platform exposure to first-time-fraud.
- Escrow timing (Section 3.4). Even Tier 2 organisers don't see most of the money until 24 hours after the event ends. Pulling a no-show stunt means losing access to the funds.
- Pre-event verification calls. For events selling >K100,000 in tickets, a platform staffer calls the organiser 48 hours before the event to confirm logistics. Five minutes of staff time, virtually eliminates the high-value fraud case. Can be automated as the platform grows — for now it's manual and worth it.

7.3 Disputes and chargebacks

When something goes wrong — event was disappointing, advertised performer didn't show, venue oversold and turned people away — the dispute flow needs an event-shaped path:

- Disputes can be opened up to 7 days after the event end-time.
- Photo / video evidence is mandatory for "event was misrepresented" disputes.
- If the organiser doesn't respond within 72 hours, the dispute auto-resolves in the buyer's favour and the platform refunds from the held escrow.
- Repeat-disputed organisers (3+ disputes upheld) lose ticketing privileges. Reinstatement requires Tier 2 KYC and platform interview. Aligns with Q62 escalation tiers.

8. Comprehensive Event Catalogue for Zambia

This is the events universe Convergeo's data model must absorb. Organised by category, each row maps to one of the five event types from Section 2 and notes the dominant pricing mode plus Zambian context. Phase 1 launch picks live in Section 9.

Category	Sub-category	Type	Pricing mode	Zambian context
Music & Live Performance	Concerts (Zambian artists)	Single ticketed	Multi-tier	Yo Maps, Slap Dee, Chef 187 etc. Lusaka/Copperbelt heavy. Hard-ticket fraud risk.
Music & Live Performance	Concerts (international touring)	Single ticketed	Multi-tier + early-bird	Periodic; high-value (K500-K2000+). Pre-event verification mandatory.
Music & Live Performance	Music festivals (multi-day)	Multi-day ticketed	Day pass + festival pass	Stanbic Music Festival, Lake of Stars adjacent. Multi-tier per day.
Music & Live Performance	Album launches	Single ticketed	Fixed or tiered	Often hybrid — public ticketed + private VIP.
Music & Live Performance	DJ nights & club events	Single ticketed	Fixed + table booking	Lusaka nightlife. Group/table pricing important.
Music & Live Performance	Open mic & live music nights	Recurring	Fixed	Weekly venue events. Recurring instances.
Music & Live Performance	Gospel concerts	Single ticketed	Multi-tier	Major category in Zambia. Often church-organised.
Comedy & Theatre	Stand-up comedy nights	Single ticketed	Fixed	Growing scene. Lusaka Playhouse, restaurant venues.
Comedy & Theatre	Theatre productions	Single or multi-day ticketed	Multi-tier	Lusaka Playhouse, university productions.

Category	Sub-category	Type	Pricing mode	Zambian context
Comedy & Theatre	Improv & sketch shows	Recurring	Fixed	Weekly / monthly venue series.
Sports	Football matches (domestic)	Single ticketed	Multi-tier (terrace/main/VIP)	Super League fixtures. Stadium ticketing — high volume, low price-point.
Sports	Football matches (international)	Single ticketed	Multi-tier	AFCON qualifiers, friendlies. Premium pricing tiers.
Sports	Rugby & netball fixtures	Single ticketed	Fixed	Smaller volume but loyal audiences.
Sports	Athletics & marathons	Single ticketed	Tiered (5K / 10K / half / full)	Lusaka Marathon, Victoria Falls Marathon. Race entry, not spectator.
Sports	Boxing & MMA events	Single ticketed	Multi-tier + table	Esther Phiri-era legacy; periodic big events.
Sports	Golf tournaments	Single ticketed + free spectator	Player entry + corporate sponsor	Often charity-tied. Player slots ticketed; gallery free.
Conferences & Professional	Industry conferences	Multi-day ticketed	Multi-tier (delegate / student / corporate)	Tech, finance, mining, agriculture. High price-point, low volume, B2B.
Conferences & Professional	Trade expos & exhibitions	Multi-day ticketed	Day pass + free with registration	Zambia International Trade Fair, agricultural shows.
Conferences & Professional	Networking events	Single ticketed	Fixed	Chamber of Commerce, professional bodies.
Conferences & Professional	Career fairs	Single ticketed or free RSVP	Free or low fixed	Universities, recruitment-driven.
Conferences & Professional	Investor pitch nights	Single ticketed	Fixed	Startup ecosystem; growing in Lusaka.

Category	Sub-category	Type	Pricing mode	Zambian context
Workshops & Education	Skills workshops	Recurring or single	Fixed or tiered	Photography, coding, business basics. Recurring template.
Workshops & Education	Wellness workshops	Recurring	Fixed	Yoga, breathwork, women's circles. Capacity tight.
Workshops & Education	Cooking & food classes	Single or recurring	Fixed	Restaurant-led cooking classes; ingredient cost included.
Workshops & Education	Art classes & paint-and-sip	Recurring	Fixed	Lusaka emerging category.
Workshops & Education	Language classes	Recurring	Fixed (subscription-style)	Bemba, Nyanja for foreigners; English for locals.
Workshops & Education	Children's workshops	Recurring	Fixed	Multi-sensory play, art, holiday programmes.
Workshops & Education	Religious teachings & retreats	Single or multi-day ticketed	Free RSVP or low fixed	Major category. Free RSVP common.
Cultural & Arts	Art exhibitions & openings	Free RSVP or ticketed	Free / low fixed	Lechwe Trust, Henry Tayali Art Centre, gallery openings.
Cultural & Arts	Film screenings	Recurring or single	Fixed (low)	LUCAC monthly series, indie cinema, documentary nights.
Cultural & Arts	Cultural festivals (traditional)	Multi-day ticketed	Multi-tier	Kuomboka, N'cwala, Likumbi Lya Mize. Tourism-tied.
Cultural & Arts	Poetry & spoken word	Recurring	Fixed	Niche but loyal scene.
Cultural & Arts	Book launches	Free RSVP or single ticketed	Free / low fixed	Authors, publishers.
Cultural & Arts	Photography exhibitions	Free RSVP	Free	Gallery + commercial space hybrids.

Category	Sub-category	Type	Pricing mode	Zambian context
Food & Beverage	Pop-up dinners & supper clubs	Single ticketed	Fixed (price includes meal)	Chef-driven, intimate. K300-K1500 per seat.
Food & Beverage	Wine tastings	Single ticketed	Fixed	Limited-seat. Shardonnay-style venues.
Food & Beverage	Cocktail & spirits events	Single ticketed	Fixed	Sponsor-driven often.
Food & Beverage	Food festivals & markets	Single or multi-day	Free RSVP + paid stalls	Lusaka food trucks, weekend markets.
Food & Beverage	Coffee cuppings & tastings	Single ticketed	Fixed	Specialty coffee scene growing.
Food & Beverage	Brunch & rooftop events	Single ticketed	Fixed	Cargo88-style; price includes meal.
Lifestyle & Community	Fashion shows & runways	Single ticketed	Multi-tier	Lusaka July, designer launches.
Lifestyle & Community	Thrift markets & flea markets	Single	Free RSVP + paid stalls	Growing weekend institution.
Lifestyle & Community	Fitness & sports community events	Recurring	Free RSVP or fixed	Park runs, cycling clubs.
Lifestyle & Community	Women-only events & circles	Single or recurring	Fixed	Major category. Game nights, wellness, circles.
Lifestyle & Community	Game nights & social games	Recurring	Fixed	Trivia, board games, themed nights.
Lifestyle & Community	Children's & family events	Single ticketed or free	Fixed	Holiday activities, theme parks, play centres.
Lifestyle & Community	Pet & animal events	Single ticketed	Fixed	Dog shows, adoption days. Niche but loyal.
Religious & Spiritual	Major religious gatherings	Single ticketed or free RSVP	Free / low fixed	Crusades, conferences. Often free RSVP at scale.

Category	Sub-category	Type	Pricing mode	Zambian context
Religious & Spiritual	Worship nights & concerts	Single ticketed	Fixed	Gospel concerts overlap; standalone worship events.
Religious & Spiritual	Retreats & camps	Multi-day ticketed	Fixed (includes lodging)	Church youth camps, women's retreats.
Markets & Expos	Agricultural shows	Multi-day ticketed	Day pass	Lusaka Show, regional shows. Annual flagships.
Markets & Expos	Trade fairs	Multi-day ticketed	Day pass + corporate	ZITF (Bulawayo) draws Zambian attendees; domestic equivalents.
Markets & Expos	Real estate & property expos	Single ticketed or free	Free / low fixed	Developer-led, sales-funnel events.
Markets & Expos	Wedding expos	Single ticketed	Low fixed	Bridal industry showcases. Vendor-funded.
Tourism-Adjacent	Adventure activity bookings	Recurring (daily)	Fixed per activity	Bungee, rafting, helicopter tours — Livingstone-centric. Overlaps with services.
Tourism-Adjacent	Safari sundowner events	Recurring	Fixed	Lodge-organised.
Tourism-Adjacent	Boat cruises & sundowner cruises	Recurring	Fixed	Zambezi sunset cruises. Capacity-tight.
Tourism-Adjacent	Wildlife & park-led events	Single or recurring	Fixed	DNPW-organised. Edutainment.
Private	Weddings (with ticketed seating)	Private	Group/table or fixed	Listed-but-unindexed. Access by code.
Private	Corporate launches & parties	Private	Free RSVP or fixed	Internal distribution.

Category	Sub-category	Type	Pricing mode	Zambian context
Private	Private parties (kitchen parties, etc.)	Private	Fixed or group	Massive Zambian category. Privacy-sensitive.
Private	Funerals (ticketed seating)	Private	Free RSVP	Coordination tool, not commerce. Sensitive UX.

That's roughly 65 event sub-categories across 11 top-level categories. Combined with the ~110 service sub-categories and ~100 product sub-categories, Convergeo's catalogue now spans 275+ supply types. The architectural payoff: 5 product classes + 6 service archetypes + 5 event types = 16 behavioural patterns, all sharing one schema, one search layer, one trust system, one payment rail. That is a genuinely unified super-app, not three apps stitched together.

9. Phased Events Launch (Aligned to the 60-Day Roadmap)

Day 49–53 of your roadmap allocates the build window for events and ticketing. With launch on Day 60, that gives you about 4 working weeks between events going live in code and the public launch. Phase 1 should target a small number of high-quality events rather than blanketing the calendar.

Phase 1 — Launch (Days 49–60 build, weeks post-launch)

Six categories where supply is achievable in a few weeks of organiser outreach and where ticketing is genuinely valuable to organisers (i.e. they're currently using WhatsApp + bank transfers and hating it):

- Workshops & education — wellness, skills, cooking, art classes. Small capacity, organiser-friendly, recurring — the perfect starter category. Recruit 5–10 facilitators.
- Comedy & theatre — defined venues, established organisers, manageable scale. Recruit 2–3 venue partnerships.
- Pop-up dinners and food events — chef-driven, intimate, currently chaotic on WhatsApp. Recruit 3–5 supper-club operators.
- Cultural & arts — gallery openings, film screenings, book launches. Mostly free RSVP at first, which sidesteps payment risk while building event-volume credibility.
- Lifestyle & community — game nights, women-only events, fitness community runs. Cheap to organise, frequent, build platform habit.
- Free RSVP events broadly — make it free for organisers in Phase 1 (no platform commission on free events ever, by design). This populates the calendar fast and gives the discovery layer something to show.

Phase 2 — Commerce deepening (Months 3–6 post-launch)

- Music concerts (Zambian artists) — the big-ticket category. Wait until trust is established and Tier 2 organiser onboarding is smooth.
- Conferences & expos — high-value, B2B, longer sales cycles. Needs the multi-tier and group-pricing UI fully shipped and tested.
- Sports fixtures — partnerships with FAZ, clubs. Stadium-scale ticketing tests the QR scanner under volume.
- Fashion shows, wedding expos — vendor-funded events that bring product/services vendors onto the platform via a different door.
- Multi-day events — agricultural shows, music festivals. Tests the multi-day ticket schema in production.

Phase 3 — Long tail and high-stakes (Months 6–12)

- International touring concerts — high-value, high-risk. Pre-event verification calls become standard. Possibly the first category where Convergeo charges premium commission (8–10% of Q5's range).
- Religious gatherings at scale — large free-RSVP capacity events. Tests the system under 5,000+ attendee scans.
- Private events (weddings, funerals, corporate) — the privacy and code-access flow requires careful UX design. Worth doing well, not first.
- Cultural festivals (Kuomboka, N'cwala) — government and traditional authority partnerships. Long lead time, high reward.
- Tourism-adjacent recurring activities — connects to the city-guide AI assistant; this is where commerce + services + events fully converge.

10. Pulling It Together

Your existing decisions — dynamic-QR, all-events scope, 5–10% commission, the Day 49–53 build window — are correct foundations. This brief sharpens the model: how events are typed, how tickets are priced and capacity-managed, how discovery works on a time-first axis instead of distance-first, how dynamic QR actually behaves at the door, how organisers onboard, and what to launch when.

The five things to take away:

- Events fall into five types (single ticketed, multi-day ticketed, recurring, free RSVP, private). One Event entity with type enum, plus EventInstance and TicketType children, covers all five without forking the codebase.
- Six ticket pricing modes need to coexist from day one, even if only fixed-price tier is exposed at launch. Multi-tier, time-staged, group/table, free RSVP, and pay-what-you-want each handle a real Zambian use case.
- Discovery is time-first, not distance-first. The events tab defaults to "Tonight + This Weekend". Distance and category filter on top. Sell-through pressure ("selling fast") is a uniquely event-shaped ranking signal worth surfacing.
- Dynamic QR is the door mechanic, not the brochure feature. HMAC-signed time-windows, 60-second refresh, PIN backup, offline scanner cache, first-scan-wins logic. Without these mechanics, "dynamic QR" is marketing language.
- Phase 1 picks six categories — workshops, comedy/theatre, pop-up food, cultural & arts, lifestyle community events, and free RSVP broadly — where supply is reachable, capacity is small, and organisers are currently using WhatsApp + bank transfers. Concerts, conferences, and high-value events wait for Phase 2. Religious-scale and international-touring events wait for Phase 3.

Combined with the products and services briefs, Convergeo's supply model is now fully specified: 16 behavioural patterns, 275+ sub-categories, one schema, one search, one trust layer, one payment rail. The "hub of logos" that Q72 commits to is no longer aspirational language — it's a documented architecture.

End of brief.

CONVERGEO

Product Strategy & Catalogue Architecture

How the platform should think about, model, and onboard physical and digital products across Zambia

Companion Brief to the Strategic Master Plan

Builds on: 75 strategic decisions, the canonical Product / VendorListing model, the Services brief
Prepared: April 2026

1. How This Connects to What You've Already Decided

Your Strategic Master Plan locks in a clear product foundation: the canonical Product table is the golden record for an item, the VendorListing table holds price, stock, and condition per vendor (decisions Q12, Q41, plus the shared-image insight you flagged yourself). That model is correct and will scale. This brief does not revisit it.

What this brief does is push one layer deeper. The canonical model works perfectly for a Coca-Cola 500ml. It works less perfectly for a kilo of fresh tomatoes, a 50kg bag of cement, a salaula bale, an Itel A23 Pro that may or may not be the same SKU as the one a competing vendor is selling, or a custom-tailored chitenge dress. Most of the architectural choices you've already made stay; what changes is the rules around when an item gets a canonical entry, when it doesn't, and how vendors create listings without polluting the catalogue.

> **Where this brief plugs in:** Section 4 of your Master Plan (Technical Architecture) defines the Product/VendorListing schema. This brief proposes additions — product classes, pricing modes, condition models, attribute groups — that fit inside that schema as JSON attribute groups rather than new tables. Nothing here breaks the 60-day roadmap.

The asymmetry between products and services

In the services brief I argued that services collapse into six transaction archetypes. Products are simpler in transaction shape — most are buy-now — but harder in catalogue shape, because Zambian commerce mixes branded global SKUs, branded local SKUs, generic commodities priced by weight or volume, second-hand goods with no SKU at all, custom-made items, and bulk/wholesale orders that need quote flows. A flat product table treats all of these the same, which is why so many African e-commerce attempts feel chaotic.

The fix is not more tables. It's recognising that the canonical-Product idea is a spectrum, not a binary, and designing the catalogue around five product classes that each behave differently.

2. The Five Product Classes

Every item that ever appears on Convergeo will fit into one of five classes. The class determines whether a canonical record exists, how vendors attach to it, what fields are mandatory, what the comparison view looks like, and how stock is counted.

Class	What it is	Canonical model	Pricing mode	Examples
A. Branded SKU	Manufacturer-defined products with a fixed identity (model, size, colour).	Strict canonical. One Product row, many VendorListings.	Per-unit, fixed.	iPhone 15 Pro Max, Samsung A55 5G, HP ProBook, Coca-Cola 500ml, Roma tomato sauce 410g.
B. Branded variant	Branded item with vendor-relevant variants (pack size, flavour) but still a defined SKU.	Canonical with variants. Variant table beneath product.	Per-unit per variant.	Sun Maize Meal 25kg vs 10kg vs 5kg, Mosi Lager 6-pack vs case, Vaseline 100ml vs 250ml.
C. Commodity / generic	Unbranded or weakly branded staples; quality varies; substitutable.	Loose canonical. One row per (commodity × grade × unit).	Per-kilo, per-litre, per-bag, or per-bunch.	Tomatoes, onions, dried kapenta, charcoal, river sand, cement (mostly), maize grain.
D. One-of-a-kind / unique	Each unit is genuinely different — no two are the same.	No canonical. Each listing stands alone.	Per-listing, often negotiable.	Used cars, used phones, salaula items, antiques, custom artwork, specific used furniture.
E. Made-to-order	Item is produced after the order is placed; spec is the order.	No canonical. Listing is a template; order carries the spec.	Quote or template-priced.	Tailored chitenge outfits, custom curtains, custom cakes, bespoke leather, custom signage.

Three things follow from this:

- The canonical-Product table you've designed is exactly right for Class A and Class B. That's where the comparison view, the shared images, the price-and-distance sort, and the hyper-local competition all happen — and those classes will dominate Phase 1 GMV.
- Class C breaks the pure-canonical model in a useful way. We need a canonical "commodity row" for tomatoes-per-kilo, but the photos and even the description have to be vendor-supplied because a heap of Soweto Market tomatoes genuinely doesn't look like a heap of Pick n Pay tomatoes. Section 3 covers this.
- Class D and Class E should not pollute the canonical Product table at all. They live in the same VendorListing table but with product_id = NULL and a class flag. Search still finds them via Meilisearch + pgvector; comparison views just don't apply. This is how you absorb salaula, used phones, and tailored work without making the catalogue feel like a flea market.

> **Schema implication (low cost, high payoff):** Add a product_class enum (A–E) on both Product and VendorListing. Add a nullable product_id FK on VendorListing (already implied by your model). For Class D/E listings, product_id is NULL and the listing carries its own title, description, images, and attributes. One enum field unlocks correct behaviour for all five classes without new tables.

3. Pricing Modes and Units of Sale

This is where most multi-vendor platforms in Africa quietly fail. They model price as a single ZMW number per listing. That's fine for an iPhone. It is wrong for almost everything in a market. The pricing model needs to support six modes from day one — even if the UI surfaces them progressively.

3.1 The six pricing modes

Mode	Description	When it applies
Per-unit fixed	Single price for one unit. Standard e-commerce.	Class A and B in almost all cases. Default mode.
Per-weight / per-volume	Price expressed per kg / per litre / per 100g; total computed at checkout from weight chosen.	Butcheries, fresh produce by kilo, bulk dried goods, fuel, paint.
Per-bunch / per-pile / per-bundle	The Zambian market reality. "K20 a heap" of tomatoes, "K10 a bunch" of rape.	Soweto, Chisokone, City Market sellers. Class C.
Tiered / volume	Different prices for different quantities (1–10 = K50, 11–50 = K45, 50+ = K40).	Wholesale-friendly listings, building materials, agro-inputs. Q37.
Range / from	"From K850" — actual price depends on customisation captured at checkout.	Class E made-to-order items. Tailoring, custom cakes, signage.
Quote-only	No public price; customer requests, vendor responds.	Bulk corporate orders, large-quantity wholesale, B2B (Q37).

3.2 Unit-of-sale matters more than you'd think

A vendor selling 50kg bags of mealie meal and a vendor selling 5kg bags should not appear on the same canonical row. They appear on linked variant rows under the same parent product. The customer sees both, can sort by price-per-kg (which the platform computes for them — vendors are not asked to do unit math), and can filter to a preferred pack size. Without that price-per-kg normalisation, comparison shopping breaks.

Therefore every product (Class A, B, C) carries two unit fields: a sale_unit (the unit the customer actually buys: bag, bottle, kg, bunch) and an optional base_unit (the comparable unit: kg, litre, gram). Where both exist, the platform always shows price-per-base-unit alongside the sticker price. This is a small piece of UX that will quietly differentiate Convergeo from every WhatsApp seller in the country.

3.3 Currency display vs. settlement

Your decision Q23 is right — display in any currency, settle in ZMW. Two refinements specific to products:

- For imported goods (electronics, cosmetics, vehicles) vendors should be able to peg the listing to USD with a daily-refreshed FX margin. The customer sees ZMW; the vendor isn't bankrupted by a kwacha slide between listing and sale.
- Lock the FX rate at order placement, not at fulfilment. Otherwise a 48-hour escrow becomes a vendor's currency-exposure problem.

4. Condition, Authenticity, and the Used-Goods Problem

Used and refurbished goods are too big a category to ignore in Zambia. Used Japan-import cars, second-hand electronics from Dubai routes, refurbished laptops, and salaula clothing are all major real categories. They're also the largest source of buyer disputes if not modelled carefully.

4.1 Condition tiers

Every Class A, B, and D listing must declare a condition from a controlled enum:

- New — sealed, with manufacturer warranty, original packaging.
- New (open box) — opened but unused, full warranty.
- Refurbished — professionally restored, working, may carry vendor warranty.
- Used — Excellent / Good / Fair — three tiers with clear platform-defined criteria, not vendor self-description.
- For parts / not working — buyer accepts the item is non-functional.

The enum matters because search filtering, return-policy logic, and escrow hold periods can all branch on it. Used items, for instance, get a slightly longer escrow window (72 hours rather than 48) because verification on arrival takes longer.

4.2 Required evidence by condition

For anything not New, the platform should require photo evidence at listing time:

- Used phones / electronics: photo of the device powered on, plus serial number / IMEI photo (the IMEI also lets the platform check against blacklisted-stolen-device databases later).
- Used vehicles: VIN photo, mileage photo, plus four exterior angles.
- Salaula: photo of the actual item being sold, not a stock photo. This is the single biggest fraud vector and is worth enforcing strictly.

4.3 Authenticity for Class A branded SKUs

Counterfeits — particularly cosmetics, phone accessories, and pharmaceuticals — are a real risk. Three layers of defence, in order of effort:

- Tier 2/3 vendors only on high-counterfeit-risk categories. Don't let Tier 1 (NRC-only) vendors list cosmetics, supplements, or pharmaceuticals.
- Brand-protection program. Once a brand has 5+ vendors on the platform, allow the brand owner (or their authorised distributor) to claim the canonical Product page and flag suspicious listings.
- Customer-side authenticity reporting. A one-tap "this looks fake" button on every listing, routed to admin queue. Cheap, effective.

5. Inventory: How Stock Should Actually Be Counted

Your existing schema has stock_quantity as an integer on VendorListing. Correct for Class A and most B. Three additions handle the rest of the catalogue.

5.1 Stock modes

- Tracked numeric — the default. "I have 7 of these". Decrements on order, blocks at zero.
- Made-to-order — Class E. Stock is effectively infinite within capacity; the constraint is lead time, not quantity. Vendor sets a per-week capacity and a default lead-time; orders past capacity queue or get rejected.
- By-weight bulk — Class C bulk goods. Vendor declares total available weight (e.g. 200kg of dried beans) and the system decrements as customers buy partial weights.
- Always-available service-like — for digital goods, infinite-supply downloads, certain rentals.

5.2 Reservation, not just decrement

During checkout, stock should be reserved for a short window (10–15 minutes is standard) rather than decremented immediately. This is in your roadmap ("Stock reservation during checkout (timeout-based release)" — day 23-24). Worth flagging the rule that goes with it: reserved stock counts as out-of-stock for new buyers, but if the reservation expires unpaid, it must release atomically. Race conditions here are how oversells happen.

5.3 Multi-vendor inventory honesty

Consider this risk: a vendor lists 50 units, sells 30 on Convergeo and 25 face-to-face at their physical shop, and Convergeo only sees 20 of those 30 sold (they marked 10 as fulfilled outside the platform). The platform now thinks they have 20 left when they have -5. Buyers order, vendor cancels, dispute count climbs.

Two mitigations, both lightweight:

- Flag vendors whose cancel-rate exceeds 5% of orders. Auto-suspend at 10%. The first is a warning shot; the second is a circuit-breaker.
- Phase 3 feature: optional POS-light integration. A vendor with a physical shop can mark off-platform sales in their dashboard with one tap, keeping Convergeo's stock count honest. Pair this with a tiny incentive — a 0.5% commission discount for vendors who do — and adoption follows.

6. How Vendors Actually Add Products

Different vendor types onboard products differently. One-size-fits-all listing forms are why platform vendor-onboarding stalls in week three. Five flows cover the universe; each maps cleanly to your existing tiered KYC and to the product classes.

6.1 Search-and-attach (the default for Class A and B)

Vendor types the product name. Convergeo's catalogue search finds the canonical Product. Vendor sees "Add your offer" and provides only price, stock, condition. Whole flow under 30 seconds. This is what your admin team curates the canonical catalogue for, and it is the single most powerful vendor-acquisition mechanic on the platform.

6.2 Submit-new-canonical (the long tail of Class A and B)

Vendor cannot find their product. They submit a candidate canonical record: name, brand, category, description, images, specs. It enters a moderation queue. Admin or AI-assisted moderation either approves it as a new canonical Product, merges it with an existing one, or rejects with reason. Vendor's listing goes live attached to whichever canonical record it ends up on.

Two practical rules around this queue:

- Auto-approve obvious duplicates of high-confidence brand+model matches. Don't make admin review every Coke 500ml submission.
- Pay a small reward (free month of Bronze, K50 credit) to vendors whose submissions become canonicals adopted by 3+ other vendors. They are doing the platform's catalogue work for free; recognise it.

6.3 Commodity quick-list (Class C)

Market traders aren't going to fill out a 12-field form. The flow is: pick category (vegetables / fish / charcoal / etc.), pick the commodity (tomatoes, kapenta, etc.) from a curated list, choose unit (per heap / per kg / per bunch), set price, snap one photo, publish. Three taps and a photo. The canonical commodity row carries category, unit options, and a generic description; the vendor's listing carries price, photo, and stock.

6.4 Unique-item listing (Class D)

Used phones, used cars, salaula. Full vendor-defined listing: title, description, multi-photo upload (mandatory: real photos of this exact item), condition tier, evidence photos for high-risk items (IMEI, VIN, mileage). Listing does not attach to any canonical Product. Search and discovery still work because Meilisearch indexes the listing's own fields and pgvector embeds its description.

6.5 Made-to-order template (Class E)

Tailor or custom maker creates a template: "Custom chitenge two-piece — from K320, 5–7 day lead time". Template carries example photos, fabric/material options, sizing form. When customer orders, they fill the spec form (chest, waist, fabric choice, optional reference photo upload), platform converts to a normal order with the spec attached.

6.6 Bulk operations

Aligns with your Q39 (manual + CSV bulk + API). Two specifics:

- CSV import schema should match the canonical Product fields, and the importer should aggressively try to match incoming rows to existing canonicals before creating new ones. This prevents the catalogue from doubling every time a Tier 2 vendor uploads.
- API access (Tier 3) should support webhook-based stock sync, not just one-shot uploads. A vendor with their own POS shouldn't need to remember to push stock; the integration pushes on every sale.

7. Product Discovery: Where the Services Logic Already Mostly Applies

Your Strategic Master Plan correctly chose Meilisearch as the primary search engine and pgvector as the semantic-AI-ready layer. The services brief argued for hybrid retrieval (BM25 + vector embeddings) with reciprocal rank fusion, then geo and quality re-ranking. Almost all of that translates directly to products. Three product-specific additions:

7.1 Two ranking signals you don't have for services

- In-stock boost. Out-of-stock listings should not appear on result page 1 except as a last resort. This is mechanical, not editorial.
- Price competitiveness within canonical. For Class A and B, where multiple vendors sell the same thing, the platform should slightly boost listings priced at or below the median for that canonical. This rewards aggressive pricing without punishing anyone — and it teaches vendors that being competitive is rewarded by visibility, which is the right behavioural lesson.

7.2 Browse vs. search vs. ask

Three discovery modes; the homepage should give equal weight to all three rather than collapsing them:

- Browse: category tree, curated collections (Q30), trending products, new arrivals. Good for users who don't know what they want.
- Search: typed query, hybrid retrieval, faceted filters. Good for users with intent.
- Ask (Phase 3, after data accumulates): natural-language assistant. "I need a phone for my mum, simple, under K2000" returns a curated handful with reasoning. This is genuinely valuable and is where pgvector pays off — but only build it once you have the data to ground it. Q67 (AI features prioritised: descriptions → fraud → pricing) and the Bootstrap-Reality tension in your Master Plan get this right.

7.3 Hyperlocal proximity for Class C

Class C goods (fresh produce, charcoal, river sand) are extremely sensitive to distance — nobody buys tomatoes from 40km away. For commodity listings, the geo re-ranking weight should be 2–3x what it is for Class A. Implementation is just a per-class multiplier on the existing distance score; no new infrastructure.

8. Comprehensive Product Catalogue for Zambia

This is the products universe Convergeo's data model must absorb. Organised by department, with each sub-category mapped to a product class (Section 2), pricing mode (Section 3), and notes on Zambian context — including which categories tend to be import-heavy, regulator-bound, or high-fraud. Phase 1 launch picks live in Section 9.

Department	Sub-category	Class	Pricing mode	Zambian context
Food: Groceries & Staples	Mealie meal & maize products	B (variants)	Per-unit per pack size	Sun, Olympic, Antelope brands. 5/10/25/50kg packs. Foundational SKU.
Food: Groceries & Staples	Cooking oil	B (variants)	Per-unit per bottle size	Sunfoil, Cargill, Pure & Sasko. Heavily price-sensitive.
Food: Groceries & Staples	Sugar & salt	B	Per-unit	Zambia Sugar dominant locally.
Food: Groceries & Staples	Rice, beans, kapenta, dried fish	C / B	Per-kg or per-pack	Mix of branded packs and weighed bulk. Kapenta is iconic; multi-grade.
Food: Groceries & Staples	Bread, biscuits, snacks	A / B	Per-unit	Manda, Bakers Inn. Daily fresh; logistics-sensitive.
Food: Groceries & Staples	Soft drinks, juices, water	A / B	Per-unit / per-case	Coca-Cola, Castle, Kachasu products. Returnable bottles complicate stock.
Food: Groceries & Staples	Tea, coffee, beverages	A / B	Per-unit	Tata Tea, local mosi-coffee. Mostly imported.
Food: Groceries & Staples	Condiments & sauces	A / B	Per-unit	Royco, Knorr, locally bottled hot sauces.
Food: Groceries & Staples	Baby food & formula	A / B	Per-unit	ZAMRA-adjacent regulation; Tier 2/3 only.
Food: Fresh & Perishable	Fresh vegetables	C	Per-bunch / per-heap / per-kg	Soweto, City Market vendors. Very short shelf life. Same-day fulfilment only.
Food: Fresh & Perishable	Fresh fruit	C	Per-piece / per-kg	Mango, banana, paw-paw, citrus. Seasonal pricing.

Department	Sub-category	Class	Pricing mode	Zambian context
Food: Fresh & Perishable	Fresh meat (beef, goat, chicken)	C	Per-kg	Butchery cuts. Cold chain required for delivery; pickup-friendly.
Food: Fresh & Perishable	Fresh fish	C	Per-kg / per-piece	Bream, kapenta (fresh), Lake Tanganyika supply. Cold chain critical.
Food: Fresh & Perishable	Eggs	B	Per-tray / per-dozen	Tray pricing dominant.
Food: Fresh & Perishable	Dairy (milk, yoghurt, butter)	B	Per-unit per pack	Parmalat, Zammilk, Trade Kings. Cold chain.
Food: Fresh & Perishable	Frozen foods	B	Per-unit	Imported frozen chicken, chips, fish fingers.
Food: Prepared & Specialty	Bakery custom (cakes, etc.)	E	Range / from	Wedding cakes, kitchen-party cakes. Made-to-order.
Food: Prepared & Specialty	Local artisan foods	B / E	Per-unit / quote	Honey, peanut butter, jams, indigenous teas — emerging premium category.
Food: Prepared & Specialty	Restaurant takeaway items	A	Per-unit	If platform supports food delivery from Phase 2. See services brief.
Beverages (Alcoholic)	Beer (lager, dark, opaque)	A / B	Per-unit / per-case	Mosi, Castle, Chibuku. Liquor licensing applies; delivery hours restricted.
Beverages (Alcoholic)	Spirits & wine	A / B	Per-bottle	Imported spirits dominant. Tier 2/3 only; age verification at delivery mandatory.
Beverages (Alcoholic)	Local brews (kachasu, munkoyo)	C	Per-litre / per-bottle	Heavily informal; legal complications; consider Phase 3+.
Personal Care & Beauty	Soap, shower gel, lotion	A / B	Per-unit	Lifebuoy, Geisha, Vaseline. High-volume daily.

Department	Sub-category	Class	Pricing mode	Zambian context
Personal Care & Beauty	Toothpaste, deodorant	A	Per-unit	Colgate, Close-Up. Brand-loyal.
Personal Care & Beauty	Hair care (relaxers, oils, weaves)	A / B / D	Per-unit / per-piece	Dark & Lovely, Motions. Weaves often unique-listing (Class D).
Personal Care & Beauty	Makeup & cosmetics	A / B	Per-unit	High counterfeit risk. Tier 2+ vendors only. Brand-protection program needed.
Personal Care & Beauty	Fragrances	A	Per-unit	Counterfeit-heavy; same controls.
Personal Care & Beauty	Feminine hygiene	A / B	Per-unit	Always, Cottex. Sensitive packaging considerations.
Personal Care & Beauty	Baby care (diapers, wipes)	A / B	Per-unit	Pampers, Huggies. Bulk discount common.
Health & Wellness	OTC medications	A	Per-unit	ZAMRA-licensed vendors only. Hard restriction on Tier 1. Prescription items not on platform.
Health & Wellness	Vitamins & supplements	A	Per-unit	ZAMRA notification required. Counterfeit-watch.
Health & Wellness	Medical devices (BP monitors, etc.)	A	Per-unit	ZAMRA / HPCZ overlap. Tier 2+.
Health & Wellness	First aid & bandages	A / B	Per-unit	Lower regulation; more permissive.
Fashion & Apparel	New clothing (men, women, kids)	A / B	Per-unit per size	Imported retail (Mr Price, Pep, Edgars-style). Size variants critical.
Fashion & Apparel	Chitenge fabric & ready-made	B / D	Per-piece / per-metre	ZamPrint and others. Fabric per-metre, ready-made per-piece. Iconic Zambian category.

Department	Sub-category	Class	Pricing mode	Zambian context
Fashion & Apparel	Tailored / made-to-order outfits	E	Range / from	Custom chitenge, suits, wedding outfits. See Section 6.5.
Fashion & Apparel	Salaula (second-hand clothing)	D	Per-piece	Massive informal category. Each item unique. Phase 2 onboarding.
Fashion & Apparel	Footwear	A / B / D	Per-pair per size	New (Class A/B), used (Class D).
Fashion & Apparel	Bags & accessories	A / B / D	Per-unit	Local leather goods are a real artisanal category.
Fashion & Apparel	Jewellery (costume + fine)	A / B / D	Per-unit	Fine jewellery: high fraud, Tier 2+ only, authentication evidence.
Electronics & Tech	Smartphones (new)	A / B	Per-unit per variant	Itel, Tecno dominate budget. Samsung, iPhone premium. Variants by storage/colour.
Electronics & Tech	Smartphones (used / refurb)	D	Per-unit	Huge category. IMEI evidence mandatory. Refurb tier.
Electronics & Tech	Phone accessories (cases, chargers)	A / B	Per-unit	Counterfeit-heavy; brand-protection useful.
Electronics & Tech	Laptops & desktops	A / D	Per-unit	New retail + huge used/import market from Dubai routes.
Electronics & Tech	TVs & home audio	A / B	Per-unit	Hisense, Samsung, LG dominant.
Electronics & Tech	Small home appliances	A / B	Per-unit	Kettles, irons, fans. Branded SKUs.
Electronics & Tech	Solar panels & inverters	A / B	Per-unit / per-kit	Booming category given load-shedding. ERB regulation for installers (services side).

Department	Sub-category	Class	Pricing mode	Zambian context
Electronics & Tech	Generators & UPS	A / B	Per-unit	Same driver as solar. Honda, Lutian, brand-name + generic.
Electronics & Tech	Gaming & cameras	A	Per-unit	Smaller market but high-margin. Counterfeit-watch on cameras.
Home & Living	Furniture (new)	A / B / E	Per-unit / quote	Home-grown carpenters (Class E) + imported branded (Class A/B). Heavy-logistics.
Home & Living	Furniture (used)	D	Per-unit	Each piece unique.
Home & Living	Mattresses & bedding	A / B	Per-unit per size	Vitafoam-style local + imports.
Home & Living	Cookware & kitchenware	A / B	Per-unit	Branded (Tefal, etc.) + generic enamel ware which is Class B/C.
Home & Living	Cleaning supplies	A / B	Per-unit	Boom, Sunlight, Surf. High-volume.
Home & Living	Curtains, mats, decor	B / D / E	Per-unit / quote	Mix of off-the-shelf and made-to-order.
Home & Living	Garden & outdoor	A / B	Per-unit	Tools, seeds for home gardens, planters.
Building & Hardware	Cement	B	Per-bag (50kg standard)	Lafarge / Mpande / Sino. Bulk discounts essential. Heavy logistics.
Building & Hardware	Iron sheets, roofing	B / C	Per-sheet / per-metre	Pricing per sheet by gauge. Bulk B2B core.
Building & Hardware	Steel rebar, wire mesh	C	Per-kg / per-piece	Construction-grade. Quote-friendly for large orders.
Building & Hardware	Bricks, blocks, sand, stone	C	Per-thousand / per-tonne	Hyper-local; transport cost dominates.
Building & Hardware	Paint	B	Per-litre per pack size	Plascon, Crown. Colour variants matter.

Department	Sub-category	Class	Pricing mode	Zambian context
Building & Hardware	Plumbing fittings	A / B	Per-unit	Pipes by length (Class C-ish).
Building & Hardware	Electrical fittings & cable	A / B / C	Per-unit / per-metre	Cable per metre is Class C.
Building & Hardware	Tools (hand & power)	A / B	Per-unit	Bosch, Stanley + cheap imports.
Building & Hardware	Doors, windows, frames	B / E	Per-unit / quote	Standard sizes Class B; custom Class E.
Automotive Parts & Supplies	New auto parts	A / B	Per-unit	OEM and aftermarket. Critical to surface compatibility data.
Automotive Parts & Supplies	Used auto parts	D	Per-unit	Each piece unique. Photos + condition mandatory.
Automotive Parts & Supplies	Tyres & batteries	A / B	Per-unit per size	Size variants critical (R14, R15, R16, etc.).
Automotive Parts & Supplies	Engine oil & fluids	A / B	Per-unit per pack	Castrol, Total. Counterfeit risk.
Automotive Parts & Supplies	Used vehicles	D	Per-unit (negotiable)	Japan-import dealers. Each car unique. VIN + mileage evidence mandatory.
Automotive Parts & Supplies	Vehicle accessories	A / B / D	Per-unit	Stereos, mats, tints — mix.
Agriculture & Inputs	Hybrid maize seed	A / B	Per-pack	Pannar, Seed Co, Zamseed. FISP-relevant. Seasonal demand spike.
Agriculture & Inputs	Vegetable & legume seed	A / B	Per-pack	Smaller volume; more variety.
Agriculture & Inputs	Fertiliser (D-Compound, Urea)	B	Per-bag (50kg)	Critical staple. Heavy logistics. Subsidy-aware pricing.
Agriculture & Inputs	Pesticides & herbicides	A / B	Per-unit	ZEMA registration. Tier 2+ only.

Department	Sub-category	Class	Pricing mode	Zambian context
Agriculture & Inputs	Animal feed	B	Per-bag	Layers mash, broiler starter, pig pellets. Volume category.
Agriculture & Inputs	Day-old chicks & livestock	B / D	Per-bird / per-head	Live animals. Vet certification. Local pickup only.
Agriculture & Inputs	Veterinary medicines	A / B	Per-unit	Vet Council regulation.
Agriculture & Inputs	Farm tools (manual)	A / B	Per-unit	Hoes, ploughs, machetes.
Agriculture & Inputs	Irrigation kits & pumps	A / B	Per-unit / per-kit	Solar irrigation booming.
Office & Stationery	Stationery (pens, paper, files)	A / B	Per-unit / per-pack	School-season demand spike.
Office & Stationery	Printers & cartridges	A / B	Per-unit	Compatible-cartridge fraud common; brand controls.
Office & Stationery	Office furniture	A / B / E	Per-unit / quote	Bulk B2B opportunity.
Office & Stationery	School supplies	A / B	Per-unit / per-pack	Heavy seasonal.
Office & Stationery	Books (textbooks, novels)	A / D	Per-unit	Used textbooks Class D.
Toys, Hobby & Sport	Children's toys	A / B	Per-unit	Mostly imported.
Toys, Hobby & Sport	Sports equipment	A / B	Per-unit	Football, netball gear. Branded + generic.
Toys, Hobby & Sport	Bicycles & scooters	A / B / D	Per-unit	New + used. Used: condition photos mandatory.
Toys, Hobby & Sport	Musical instruments	A / D	Per-unit	Heavy used market.
Toys, Hobby & Sport	Art & craft supplies	A / B	Per-unit	Smaller niche.
Crafts & Cultural	Curios, carvings, baskets	D	Per-piece	Tourist-driven. Each piece unique. Class D listing.
Crafts & Cultural	Pottery & ceramics	D	Per-piece	Local artisans.
Crafts & Cultural	Beadwork & jewellery	D / B	Per-piece	Mix of one-off (D) and small-batch (B).

Department	Sub-category	Class	Pricing mode	Zambian context
Crafts & Cultural	Local art / paintings	D	Per-piece	Each unique.
Pets	Pet food	A / B	Per-unit per pack	Royal Canin, IAMS premium; local mass-market.
Pets	Pet accessories	A / B / D	Per-unit	Collars, beds, toys.
Pets	Live pets	D	Per-animal	Vet certification. Local pickup only. Phase 3+.
Digital Goods	E-vouchers & gift cards	A	Per-unit (denomination)	Mobile money top-up, retail vouchers.
Digital Goods	Software licenses	A	Per-unit	Niche but feasible.
Digital Goods	Event tickets	A	Per-unit (tier-priced)	Already in your roadmap; dynamic QR (Q50). Sits in product taxonomy with class=ticket flag.
Digital Goods	Online courses & e-books	A	Per-unit	Expand from Phase 3.

That's roughly 100 product sub-categories across 13 departments. Combined with the ~110 service sub-categories from the previous brief, Convergeo's data model needs to cleanly absorb 200+ supply types from across both halves of the platform. The five product classes plus the six service archetypes give you exactly that — eleven behavioural patterns, one underlying schema.

9. Phased Product Launch (Aligned to the 60-Day Roadmap)

Your roadmap reaches public launch on Day 60 with a target of 75–100 active vendors and at least 5 products each. This section maps which product departments those first vendors should come from, and which categories wait.

Phase 1 — Launch (Days 1–60, weeks 6–8 onboarding)

Eight product departments. Pick vendors who already have stable supply, photograph well, and can fulfil reliably. The goal is depth in a few areas, not thinness across all.

- Food: Groceries & Staples — mealie meal, oil, rice, sugar, beverages. Highest repeat-purchase frequency, builds habit.
- Personal Care & Beauty — soap, lotion, hair care. Tier 2+ for makeup; skip fragrances at launch.
- Fashion & Apparel — chitenge fabric and ready-made; new clothing from formal retailers. Skip salaula at launch (Class D complexity).
- Electronics & Tech (selective) — new phones (Itel, Tecno), accessories, small appliances, solar kits. Skip used phones at launch.
- Home & Living — kitchenware, cleaning supplies, branded furniture. Skip used furniture and made-to-order.
- Office & Stationery — pens, paper, school supplies. Easy supply, good for school-season campaign.
- Building & Hardware (light) — paint, hand tools, small fittings. Skip cement / sand / heavy categories at launch (logistics not ready).
- Event tickets — already in your Day 49–53 plan.

Phase 2 — Commerce deepening (Months 3–6 post-launch)

Add classes that need more catalogue maturity, more vendor trust history, or heavier logistics:

- Class C produce (fresh vegetables, fruit, fish, meat) — needs same-day fulfilment. Pilot in Lusaka first.
- Class D used phones, used laptops, used furniture, salaula — needs the condition + evidence + dispute-handling system fully battle-tested.
- Heavy building materials — cement, iron sheets, sand, blocks. Needs vendor-managed delivery (your Q45 hybrid logistics) and bulk-pricing UI.
- Agriculture inputs — seasonal but huge: time the rollout to the September–November planting window.

- Used vehicles — VIN/mileage evidence flow, longer escrow, possibly even-longer dispute window.

Phase 3 — Long tail and specialised (Months 6–12)

- Class E made-to-order — tailoring, custom cakes, custom signage. Template flow + spec form.
- Pharmaceuticals (OTC), supplements — only with ZAMRA-licensed vendors and authentication program.
- Crafts & cultural curios — feeds the tourism/city-guides angle (Q52).
- Live animals, alcoholic beverages with delivery, regulated specialty goods — last because regulation cost is highest.
- AI assistant for product discovery ("ask" mode) — once you have ≥10K transactions to ground recommendations on.

10. Pulling It Together

Your existing decisions handle the e-commerce mechanics — escrow, payments, ranking, search, comparison. This brief sharpens the catalogue itself: how products are classified, how they're priced, how stock is counted, how vendors add them, and which to launch when.

The five things to take away:

- Products fall into five classes (A: branded SKU, B: branded variant, C: commodity, D: unique, E: made-to-order). Your canonical Product / VendorListing schema works for all five if you add a product_class enum and allow product_id to be NULL for D and E.
- Pricing has six modes, not one. Per-weight, per-bunch, tiered, range, and quote-only have to coexist with per-unit fixed from day one — even if only per-unit is exposed in the UI at launch.
- Condition is a controlled enum with evidence requirements. This is what makes used goods (a huge Zambian category) safe to transact and what makes the dispute system tractable.
- Five vendor onboarding flows map to the five classes. Search-and-attach is the default and is the platform's largest vendor-acquisition lever — protect it by curating a strong canonical catalogue early.
- Phase 1 picks eight product departments where supply is stable and logistics are simple. Class C, D, and E categories — and the messier regulated ones — wait until the trust and operations layers are battle-tested.

Combined with the services brief, Convergeo now has a coherent model for absorbing essentially any kind of supply Zambia produces: 11 behavioural patterns (5 product classes + 6 service archetypes), 200+ sub-categories, one schema, one search layer, one trust system. That is the actual moat.

End of brief.

CONVERGEO

Business, Pipelines & Three-Frontend Architecture

How Convergeo serves businesses, runs B2C and B2B side by side, and moves stock from factory to customer

Companion Brief to the Strategic Master Plan

Builds on: 75 strategic decisions, the Services / Products / Events briefs
Benchmark: Alibaba (1688 / Taobao / Tmall / Alibaba.com) + Alipay
Prepared: April 2026

1. The Alibaba/Alipay Benchmark, Honestly Read

Your vision is to build something with the service-delivery efficiency of Alipay and the multi-modal commerce reach of Alibaba (including its AI-mode features). Before architecting toward that benchmark, it's worth being precise about what Alibaba and Alipay actually are, because the surface-level reading misleads.

1.1 Alibaba is not one platform. It's four.

Alibaba Group operates four separate marketplaces, each with its own frontend, brand, audience, and commercial mechanics:

- 1688.com — domestic Chinese B2B wholesale. Manufacturers and large trading companies sell in bulk to resellers, retailers, and SMEs. RFQ-driven, MOQ-heavy, factory-direct pricing. Mostly Chinese-language.
- Taobao.com — C2C and small-business B2C retail, individual consumers. "The eBay/Amazon of China." No transaction commission; revenue from advertising. Massive product variety.
- Tmall.com — premium B2C for established brands and registered businesses. Strict quality control, brand-protection, where Gucci and Burberry sell to Chinese consumers. Pay-to-play.
- Alibaba.com — international B2B. Connects Chinese manufacturers and exporters to global business buyers. English support, Trade Assurance, cross-border logistics.

Each frontend is purpose-built for one transaction shape. They share underlying infrastructure (Alipay, Cainiao logistics, account systems), but the user-facing apps are separate. Jack Ma's quote, paraphrased, is the design principle: "When you try to please everyone you please no one."

1.2 Alipay is not a payment processor. It's a financial OS.

Alipay started as escrow for Taobao buyers ("I'll trust the seller if Alipay holds my money") and evolved into a super-app spanning payments, wallet, transfers, lending, insurance, wealth management, identity, public services, and merchant tools. The escrow heritage is what made it trusted; the super-app expansion is what made it ubiquitous. Convergeo's escrow + mobile-money + commission engine (Q19–Q22 in your Master Plan) is the same starting move.

1.3 What this means for Convergeo at a $2K bootstrap budget

Building four separate platforms is impossible at your budget and timeline. But the principle behind the four-platform model — different transaction shapes need different frontends — still applies. The right Zambian adaptation is:

- Three frontends, not four. B2C (the customer app, your Q34 PWA), B2B Vendor (the seller dashboard), and Admin. This is already in your Master Plan as the Customer / Vendor / Admin trio (Q12).

- B2B and B2C inside one platform via a mode-toggle on the buyer side. A registered business buyer at a Tier 2/3 vendor can flip into B2B mode — wholesale prices visible, MOQs enforced, RFQ enabled, Net-30 terms available — without leaving the app. A consumer never sees those mechanics.
- Vendors can simultaneously be sellers and B2B buyers. The bakery owner sells finished goods on the consumer side and buys flour, sugar, packaging from wholesale vendors on the B2B side. One account, two hats.

This is the Alibaba four-platform vision compressed into one platform with role-aware UI. It's the only architecture that fits a $2K bootstrap and still scales to the super-app you want over five years.

> **The Alipay-shaped insight worth internalising:** Alipay didn't try to be everything on day one. It became indispensable by being the most trusted way to settle a Taobao transaction, then expanded outward as that trust accumulated. Convergeo's path is the same: be the most trusted way to settle a transaction in Zambia, then layer financial services as the user base earns the right. Don't pre-build the wallet, the lending, the insurance. Earn them.

2. The Six Business Archetypes Convergeo Must Serve

Convergeo's vendor base is not homogeneous. A market trader and a beverage manufacturer are both "vendors" in the schema, but everything else about them — what they sell, how they price, who buys, how they fulfil, what tools they need — diverges sharply. Six archetypes cover the universe; each has a tailored experience inside the same Vendor frontend, driven by what they declared during onboarding.

Archetype	Description	What they need from Convergeo
1. Market Trader / Informal SME	Sole proprietor, NRC-only KYC (Tier 1, Q36), 1–50 product catalogue, often single location.	Three-tap product listing, mobile-first dashboard, instant payouts, Bemba/Nyanja UI (Q27). Minimal analytics. WhatsApp order alerts.
2. Registered Retailer	PACRA-registered (Tier 2), 50–500 SKUs, one or more shops. Sells finished goods.	Catalogue management, multi-branch inventory, CSV import, basic analytics, verified badge. Bronze/Silver tier features (Master Plan §3).
3. Service Professional	Plumber, photographer, lawyer, salon owner. May be Tier 1 or Tier 2 depending on scale.	Calendar, RFQ inbox, quote builder, deposit-and-balance payments, portfolio gallery, response-time tracking.
4. Manufacturer / Producer	Makes goods at scale. Local food processors, textile mills, furniture workshops. Tier 2 minimum.	Bulk listing tools, MOQ management, B2B-only pricing tier, distributor channel management, production lead-time fields, dispatch scheduling.
5. Importer / Wholesaler / Distributor	Imports or wholesales to other businesses. Tier 2/3. Often inventory-heavy with warehouse stock.	Tiered pricing engine, warehouse-aware inventory, B2B credit terms, account-manager assignment, bulk-order RFQ inbox, ERP-style API (Q39 Tier 3).
6. Event Organiser / Venue	Concert promoters, conference organisers, wedding venues, workshop facilitators.	Event creation wizard, ticket-tier builder, dynamic-QR scanner, attendee dashboard, promo codes, co-organiser roles. (Covered fully in the Events brief.)

The onboarding wizard's first question is "What kind of business are you?" — and the answer wires the dashboard. A Market Trader never sees an MOQ field; a Manufacturer never sees a calendar; an Event Organiser never sees a CSV importer. Same Vendor frontend, six profiles.

3. How Convergeo Supports Each Business Across All Four Supply Types

A business may sell products, services, events, and inventory at the same time. A bakery sells cakes (products), runs Saturday baking workshops (events), offers custom wedding-cake commissions (services), and may run a wholesale arm supplying restaurants (B2B inventory). Convergeo treats these as distinct supply types living under one Vendor account, and provides specific tooling for each.

3.1 Product support

For every business that sells physical or digital goods (Archetypes 1, 2, 4, 5), the platform provides:

- The five product classes (A–E) and six pricing modes from the Products brief — branded SKUs to commodities to made-to-order.
- Search-and-attach to the canonical catalogue, so a vendor selling Coke 500ml never has to write that listing — they just attach an offer to the master record.
- Stock tracking with reservation logic, multi-location inventory (per branch), low-stock alerts, and optional POS-light off-platform sales tracking.
- Image pipeline through Cloudinary (Q16) — vendors upload once; platform handles compression, responsive sizing, lazy loading.
- Variant management — sizes, colours, pack sizes — with their own price and stock per variant.
- Bulk update tools — CSV upload (Tier 2), API webhooks for stock sync (Tier 3, Q39), bulk price changes via the dashboard.

3.2 Service support

For service-providing vendors (Archetype 3 primarily, but extending into 2 and 6), the platform provides:

- The six service archetypes from the Services brief — Buy-now, Bookable, Reservation, Walk-in, Quote-based, Event ticket.
- Calendar with availability rules, recurring slots, blackout dates, multi-staff assignment for salon/clinic-style vendors.
- RFQ inbox for quote-based services (Q51) — "3–5 providers usually reply in an hour". Customer broadcasts a job; vendor responds with a quote.
- Deposit + balance payments for bookable services where a deposit secures the slot and the balance is due on completion.
- Portfolio gallery — before/after photos for trades, completed work for creatives, video introductions for premium tiers.
- Response-time tracking, visibly displayed on the vendor's profile ("usually replies within 2 hours"). Single highest-leverage trust signal for new service providers.

3.3 Event support

For event-running vendors (Archetype 6, plus extensions when retailers run sales events or workshops), the platform provides:

- The five event types from the Events brief — single ticketed, multi-day, recurring, free RSVP, private.
- Event creation wizard, three-screen flow, save-as-draft, multi-tier ticket builder.
- Dynamic QR ticketing (Q50), PIN backup (Q33), offline scanner cache.
- Co-organiser roles (Owner / Manager / Door) scoped per event.
- Promotion tools — auto-generated shareable link with rich Open Graph preview, promo codes, affiliate links, post-event email/SMS to attendees.
- Live attendance dashboard during the event.

3.4 Inventory support for businesses that hold stock at scale

This is the layer that hasn't been covered in any prior brief and is critical for Archetypes 4 and 5. Manufacturers and importers don't just need a stock field on a product row — they need warehouse-aware inventory management.

- Multi-warehouse stock model. A vendor with depots in Lusaka, Ndola, and Kitwe sees stock per location. Customer-side, the platform routes the order to the nearest stocked depot. Out-of-stock at one location doesn't kill the listing.
- Lot and batch tracking for FMCG with expiry dates (food, pharma, cosmetics). Each batch carries production date, expiry date, lot number. FIFO dispatch by default.
- Goods-received workflow. When new stock arrives at a warehouse — production output for manufacturers, container clearance for importers — the vendor scans a packing list (or uploads a CSV) and the system increments stock with the batch metadata.
- Reorder points and forecast assist. The system suggests reorder timing based on sales velocity. Crude version: "You sold 240 units in the last 30 days; at current pace your stock runs out in 14 days." Useful version (later): demand forecasting that accounts for seasonality. The crude version is what ships in Phase 2.
- Wholesale-channel inventory pool. A manufacturer or distributor can ring-fence a portion of stock for B2B customers only — "5,000 units retail, 20,000 units B2B-only." Retail buyers can't drain wholesale inventory, and wholesale buyers see deeper stock than the consumer view shows.
- Production lead time as a first-class field. For made-to-order manufacturing (custom furniture, branded merchandise printing), the listing carries a lead-time range and the system calculates promised delivery dates from order placement, not from stock availability.

Where this lives in the schema: Add a Warehouse entity (vendor_id, location, name, GPS), an InventoryLot entity (warehouse_id, product_id or vendor_listing_id, batch_number, expiry_date, quantity_on_hand, quantity_reserved), and a wholesale_pool_quantity field on VendorListing. The ProductBatch logic plugs into the existing stock validation at checkout — checkout now picks the right batch (FIFO or earliest-expiry-first) instead of just decrementing a single integer.

4. The Three Frontends, In Detail

Q12 of your Master Plan locks in three separate frontend apps sharing one Django REST API: Customer (PWA, Q34), Vendor, and Admin. This section spells out what each frontend actually contains, what it deliberately doesn't, and how the three interlock without colliding.

4.1 The Customer (B2C) Frontend — the PWA

This is what the public sees. The mass of users, the demand side. Designed for low-friction, mobile-first, data-light consumption (your Q34 PWA decision is the right one). Every byte saved here is a conversion preserved.

Primary surfaces

- Home — search bar at top, curated collections, categories, trending, events module defaulting to "Tonight + This Weekend", vendor spotlight (Q30 hero/B/C/D layout).
- Search & browse — Meilisearch-powered (Q15) with faceted filters, sort by relevance/price/distance/rating, semantic fallback via pgvector (Q14).
- Product / service / event detail pages — image gallery, vendor info card, comparison view ("7 vendors selling this — sort by nearest / cheapest"), reviews with verified-purchase badge (Q32), add-to-cart or book-now or RFQ-now action.
- Cart & checkout — multi-vendor cart with vendor sub-totals, address or pickup-point selection (Q44), payment method picker (mobile money / card / COD per Q25), escrow visualisation (the four-step tracker that beats WhatsApp sellers).
- My orders — status timeline, QR code for pickup (Q33), reorder, dispute, review.
- Wishlist, recently viewed, saved searches — stickiness without aggression.
- Profile — addresses, payment methods, language toggle (English / Bemba / Nyanja, Q27), notification preferences, business-mode toggle (see B2B section below).
- City guides (Phase 2) — Lusaka / Livingstone / Ndola, blending products, services, and events into a single discovery surface (Q52).

What the Customer frontend deliberately doesn't have

- No inventory dashboards, no analytics, no vendor tooling. Customers don't need them and showing them dilutes the experience.
- No B2B mechanics by default. Wholesale prices, MOQs, Net-30 terms, RFQ-for-bulk are hidden until the user opts into business mode.
- No app-store install dance. PWA install banner only (Q34). Acquisition cheat code intact.

4.2 The Vendor (B2B-seller) Frontend

This is the business workbench. It is consciously less polished and more dense than the Customer frontend because vendors will spend an hour a day in it, not three minutes. Information density is a feature here, not a bug.

Mobile vs desktop split (your two-pronged workbench from the Master Plan)

- Mobile daily-driver — for market traders and small vendors running their business from a phone. Today's sales, orders needing action, a giant "mark packed" button, simple stock add. Status-at-a-glance, not analysis.
- Desktop dashboard — for Tier 2/3 vendors with back-office time. Multi-graph analytics (GMV trends, conversion, traffic sources), CSV bulk operations, multi-warehouse inventory, B2B account management, complex pricing rules. Also accessible on tablet.

Primary surfaces

- Dashboard — today's KPIs, pending actions, sales trend, recent reviews, low-stock alerts.
- Catalogue — products / services / events tabs, with the right tools surfaced for each.
- Inventory — stock by location, lot/batch, reorder suggestions, goods-received workflow, wholesale pool slider.
- Orders — incoming queue, fulfilment status, shipping, returns, disputes.
- B2B inbox — RFQs from business buyers, quote builder, account-buyer relationships.
- Customers / B2B accounts — for vendors with repeat business buyers, a CRM-light view of their B2B clients with order history, credit terms, custom pricing.
- Marketing — promo codes, featured-listing slots, social-share helpers, post-purchase email/SMS to past buyers (subscription tier and above).
- Payouts & finance — balance held in escrow, payouts scheduled, settlement history, tax-ready exports (Q24 VAT-ready architecture).
- Storefront customisation — logo, theme colour, about, social links, hours, photos, branches (the "hub of logos" promise from Q72 — this is where vendors actually configure their public face).
- Insights — page views, search impressions, conversion funnel, top search terms that landed buyers on their page (the data they can't get from anywhere else, which is what keeps them on Convergeo).

4.3 The Admin Frontend — Convergeo's command center

This is what your team and you use to operate the platform. It is not externally facing. The Master Plan's Day 35-36 already commits to the basics; this section spells out what "effectively and efficiently running the system" actually means.

Operational surfaces

- Vendor management — list, search, view detail, approve/reject/suspend. KYC document review with side-by-side comparison.
- Product moderation queue — new canonical-record submissions (from Section 6.2 of the Products brief), price-anomaly flags, counterfeit reports, approve/reject with comments.
- Order operations — order search, intervention on stuck orders, manual escrow release/refund, dispute resolution interface with vendor + buyer chat history.
- User management — search, view, suspend/ban, role assignment (admin vs moderator vs read-only).
- Content management — homepage curation (Q30 hero / collections), category tree editing, banner management, city-guide content.
- Trust & safety — fraud queue, suspicious-account flags, dispute volumes per vendor, automatic alerts when vendor cancel-rate or refund-rate exceeds thresholds (5% warning / 10% auto-suspend, per the Products brief Section 5.3).

Financial surfaces

- Platform GMV — daily, weekly, monthly. Drill-down by category, region, vendor.
- Commission accruals — variable by category per Q2 (5% electronics, 12% fashion, etc.).
- Payout queue — vendor payouts pending, escrow age, bottlenecks.
- Subscription billing — Bronze/Silver/Gold/Platinum (Q4) — who's paid, who's renewing, who's churning.
- Reconciliation — DPO and Lenco settlement reports vs internal ledger. Q19 dual-gateway means dual reconciliation.

Strategic / analytics surfaces

- Customer cohorts — acquisition by channel, retention curves, repeat-purchase rate.
- Vendor cohorts — onboarding-to-first-sale time, GMV by tier, churn signals.
- Search analytics — top searched terms, zero-result queries (which directly drives canonical-catalogue work).
- Live monitoring — error rates, payment success rates, p95 latency, uptime per service. Q63 "Solo founder + AI tools" needs this to compensate for the absent ops team.
- AI-assisted operational ops (Q63, Q66, n8n) — automated email queues, abandoned cart recovery, vendor onboarding sequences, post-purchase reviews requests, all visible and pausable from this surface.

> **The Q63 reality:** You're a solo founder. The Admin frontend isn't a luxury — it's the multiplier that makes Q63 viable. Every decision in the Admin UI should answer the question "could AI or n8n do

> this?" If yes, automate it and surface the audit log. If no, design the manual interface to take 30 seconds, not 5 minutes.

5. The B2B Layer: Alibaba/1688 Mechanics Inside the Same Platform

This is the section that turns Convergeo from "Zambian Jumia with services and events" into the Alibaba-shaped vision in your prompt. Building a separate 1688-clone alongside Taobao isn't realistic at $2K. The right move is a B2B layer that activates inside the existing Customer and Vendor frontends when both sides opt in.

5.1 Who can sell B2B

Tier 2 (PACRA-registered) and Tier 3 (premium) vendors can opt into B2B selling. Tier 1 cannot — wholesale transactions involve invoice-grade documentation, larger amounts, longer payment terms, and tax compliance that informal vendors aren't equipped for. This restriction also creates one more incentive for Tier 1 vendors to upgrade — the Master Plan's progressive-formalisation thesis applies to the supply side too.

5.2 Who can buy B2B

Any registered business — verified via PACRA upload — can flip into business-mode in the customer app. The toggle is a profile setting; once verified, the business sees:

- Wholesale price tiers on listings that have them ("Retail K50, 10+ K42, 50+ K38").
- MOQ enforcement — listings that require minimum order quantities now show them and enforce at checkout.
- RFQ button on bulk-friendly categories, not just services.
- Net-30 / Net-60 payment terms where the vendor offers them and the buyer's account has been credit-approved.
- Tax-compliant invoices automatically generated for every order, with the buyer's business details on them.
- Multi-user accounts — one business can have a procurement officer, a finance officer, and an approver, with role-based permissions on cart and checkout.

5.3 The B2B-specific mechanics

Five mechanics are B2B-specific. They do not appear in the consumer experience and they shouldn't be built until Phase 2 of the post-launch roadmap, but the data model needs space for them from day one.

- Tiered pricing engine — already in Q37. Listing carries a price_tiers array: [{min_qty: 1, price: 50}, {min_qty: 10, price: 42}, {min_qty: 50, price: 38}]. Cart calculates the right tier automatically based on quantity.

- RFQ for bulk purchases — extension of Q51's services RFQ. A B2B buyer needing 500 units of something posts a request; vendors selling that product get notified and respond with a quote and lead time. Reverse-marketplace flow.
- Net-30 / Net-60 payment terms — only available to credit-approved B2B buyers (admin reviews, Tier 2 PACRA verification + clean payment history). The platform fronts the cash to the vendor on order, then collects from the buyer on the agreed term. This is genuinely a small lending product and should not launch until volume justifies the credit risk — Year 2 at earliest.
- Account managers — for top vendors and top buyers, a Convergeo staff member is assigned. Visible in the Vendor / B2B-buyer dashboard with direct WhatsApp contact. Manual touch, automated through n8n notifications.
- Contract pricing — pre-negotiated per-buyer prices that override the public tiered prices. "Vendor X has agreed K35/unit for Buyer Y up to 10,000 units per month." Stored as a contract record, applied at cart time when both parties match.

5.4 What this looks like as a flow

Concrete example: Mwansa runs a small bakery in Kabwata. She uses Convergeo as a customer to buy flour and sugar in bulk from Mike's Mills (a Tier 2 wholesaler), and as a vendor to sell finished cakes to consumers and to a few restaurants:

- Mwansa logs into Convergeo. Her account has business-mode toggled on (PACRA-verified) and is also a vendor account. She sees both hats simultaneously: a buyer view and a seller view, switchable in the top nav.
- On the buyer side: she searches "flour", filters to wholesale pricing, sees Mike's Mills offering 50kg bags at K780 retail / K720 for 10+ / K680 for 50+. She orders 20 bags, paying via mobile money on Net-15 terms (her credit limit allows it). The order routes to Mike's Mills' Lusaka warehouse for next-day delivery.
- On the seller side: she sees today's incoming orders for cakes from consumers (escrowed, paid). She also has an RFQ from a hotel asking for 50 chocolate cakes for an event in 3 weeks. She clicks Quote, enters K150/cake with K500 setup fee, the hotel accepts, deposit is paid into escrow.
- All of this happens in one Convergeo account, one phone, one transaction layer. That is Alibaba's four-platform design compressed into Convergeo's one platform with role-aware UI.

6. The Supply-Side Pipeline: From Production to Listing-Ready

This pipeline traces a unit of stock from the moment it's produced or imported until it's available for purchase on the platform. Different archetypes (manufacturer vs importer vs retailer vs market trader) enter the pipeline at different stages, but the rails are the same.

Stage 1 — Production or Procurement Manufacturer makes goods, or importer / wholesaler procures them externally. This stage is upstream of Convergeo — the platform doesn't run factories — but the next stage's quality depends on what comes out of this one. Practical Convergeo touchpoint: production lead-time field on listings means the schema knows whether a stock unit was produced or procured externally.

Stage 2 — Goods-Received Event Stock arrives at a warehouse / shop / depot. Vendor opens the Vendor app's Inventory → Goods-Received flow. Three options: (a) scan a packing list barcode; (b) upload a CSV; (c) manual entry for small vendors. Each line creates an InventoryLot record with batch number, expiry date (if applicable), quantity, source warehouse. This is the moment stock becomes platform-visible.

Stage 3 — Listing Creation or Stock Increment If the product already has a VendorListing, the goods-received event simply increments that listing's available quantity. If not, the vendor creates a new listing — search-and-attach to canonical (Class A/B), commodity quick-list (Class C), unique-listing (Class D), or made-to-order template (Class E) per the Products brief. Listing carries the warehouse_id and lot_id so checkout knows where to draw from.

Stage 4 — Quality & Compliance Gating For regulated categories (pharma, cosmetics, food, agricultural inputs), the listing enters a moderation queue. Admin (or AI-assisted moderation per Q67) verifies the licence, checks photos for obvious counterfeits, runs price-anomaly detection. For non-regulated Class A/B listings attaching to canonical records, this is auto-approved. For Class D used items above K1,000, manual review is required for the first 5 listings of each new vendor — after that, trust accrues.

Stage 5 — Wholesale Pool Allocation If the vendor has B2B selling enabled, they specify wholesale-pool quantity at this point. Example: 5,000 units total, 3,000 retail-only, 2,000 wholesale-only. Two virtual stock buckets, one underlying physical inventory, separate availability per buyer type.

Stage 6 — Pricing & Tier Setup Vendor sets retail price and (if B2B) wholesale tier prices. Volume discounts, bulk MOQs, time-staged early-bird logic for events. Pricing rules can include automatic FX-pegged adjustments for imported goods (Section 3.3 of Products brief) — vendor lists in USD with a daily ZMW conversion margin so kwacha slides don't bankrupt them mid-listing.

Stage 7 — Distribution Decision (B2B-Specific) Manufacturer-archetype vendors can route stock to other Convergeo vendors as wholesale buyers, in addition to selling consumer-direct. A bottler can sell 1,000 cases to a chain of mini-marts on the platform, who then resell on Convergeo retail-side. The platform tracks the chain — manufacturer → distributor → retailer → consumer — for taxation and trust traceability.

Stage 8 — Discovery Activation Listing is now indexed by Meilisearch (Q15) and embedded by pgvector (Q14) for semantic search. It's eligible for category browse, search, recommendations. Curated-collection placement (Q30) is admin-controlled, not automatic — a vendor can't pay for the homepage hero in Phase 1 (auction-based promoted listings come later, per your revenue Layer 4).

Stage 9 — Inventory Health Monitoring Daily, the system checks each listing's stock-vs-velocity. Flags low-stock at 14 days remaining, suggests reorder, alerts vendor via WhatsApp / SMS. This is the simple-but-useful version; demand forecasting that handles seasonality is a Phase 3 enhancement.

That's the supply pipeline in nine stages. A market trader collapses Stages 1–5 into one moment of standing in front of their stall and snapping a photo. A manufacturer with three depots and a B2B channel touches all nine. The schema and APIs handle both.

7. The Demand-Side Pipeline: From Customer Intent to Item-In-Hand

This pipeline traces the customer's journey from opening Convergeo with an intent in mind to physically holding the item or experiencing the service. Most of these stages are in your existing 60-day roadmap; this section sequences them as one continuous flow and highlights what happens at each handoff.

Stage 1 — Discovery Customer opens the PWA, lands on home or arrives via a shared link. Three discovery modes available (per the Products brief): browse (categories, collections), search (Meilisearch + pgvector hybrid), or in Phase 3 ask (natural-language assistant). Time-first defaults for events, distance-first defaults for services, intent-first for products. The discovery layer's job is to surface the right 5–10 candidate listings.

Stage 2 — Comparison & Decision For Class A/B canonical products, the comparison view shows multiple vendor offers — sort by price / nearest / fastest delivery / rating. For services, the listing shows reviews, response time, sample work. For events, ticket tiers, capacity remaining, organiser verification. Customer adds to cart, books a slot, or initiates an RFQ. This stage is where the canonical-catalogue work pays off most visibly.

Stage 3 — Cart Build Customer assembles their order. Cart is multi-vendor (Q12 sub-orders), per-vendor sub-totals visible. Stock validated in real-time — out-of-stock items are flagged before checkout. Cart persists across login states (anonymous → authenticated merge, per the Day 16-17 spec).

Stage 4 — Address & Fulfilment Choice Customer selects delivery to a saved address, pickup from a designated point (warehouse, vendor shop, partner pickup point per Q44), or pay-at-pickup for the COD-friendly tier. Delivery fee calculated by zone (your Day 18-19 spec). For free delivery thresholds (Q6, K200+), the cart shows the customer how close they are: "Add K34 more for free delivery."

Stage 5 — Payment Customer picks payment method: mobile money (MTN, Airtel, Zamtel — Q20), card (Visa/Mastercard via DPO), or COD where allowed (Q25). For B2B users with Net terms, that option appears too. Payment is initiated; for mobile money, a USSD prompt arrives on the customer's phone and they approve. Funds land in escrow (Q22) — the customer sees the four-step trust-state visualisation that beats WhatsApp sellers.

Stage 6 — Order Confirmation & Vendor Notification Order is created (atomic transaction, stock decremented, cart cleared, multi-vendor split into sub-orders per the Day 23-24 spec). Vendor receives push notification + SMS + WhatsApp message (Q68). Vendor opens the order in their app, sees customer info, items, delivery details, and confirms or rejects with reason.

Stage 7 — Fulfilment Picking Vendor pulls stock from the right warehouse / lot (FIFO by default, batch-aware for expiry-sensitive items), packs the order, prints / displays the label or QR code. Vendor marks order as packed in their app. State machine transitions: Placed → Confirmed → Processing → Shipped (Day 25-26 spec).

Stage 8 — Logistics Handoff Order is collected by the appropriate logistics layer. Three paths per Q43/Q45: (a) platform-aggregated delivery for small standard items via Yango or local courier — vendor drops at a pickup point or courier collects from vendor; (b) vendor-managed delivery for furniture / appliances / heavy items; (c) customer pickup with QR-code verification (Q33, Q50 for events). Tracking ID synced to the customer's order page (Q47 — pass-through tracking, not built from scratch).

Stage 9 — In-Transit & Arrival Customer tracks the order through their app. WhatsApp updates at major state changes — picked up, out for delivery, arrived. Estimated arrival window communicated. For pickup orders, customer gets a notification when ready and presents QR + PIN at collection. State: Shipped → Delivered.

Stage 10 — Confirmation & Escrow Release Customer taps "Confirm Received" in their app, or — if 48 hours pass without a dispute — auto-confirmation triggers. Escrow releases to vendor (Q22). Vendor's payout schedule queues the funds for settlement. State: Delivered → Completed.

Stage 11 — Review & Reorder Customer is prompted to review (verified-purchase only, per Q32). Vendor can respond. Review feeds the discovery layer's quality score. Customer's order history populates personalised recommendations (Q69 — "recommended for you"). One-tap reorder for repeat-purchase categories like groceries — single biggest retention mechanic in FMCG e-commerce.

Stage 12 — Dispute or Refund (Exception Path) If customer is unhappy, they open a dispute within the dispute window (Day 43-44 spec). Evidence submitted, vendor responds, admin resolves if needed. Refund processes from held escrow if buyer wins. For events (per Events brief Section 5.5), the matrix is event-specific.

Twelve stages. The first seven happen in minutes, stage 8-9 take hours to days, stage 10 is the trust moment, and stage 11-12 close the loop. Convergeo's job is to make every stage feel as fast and certain as possible — and to make the failure modes survivable.

8. Where the Two Pipelines Meet

The supply and demand pipelines are not parallel — they intersect. Understanding the intersection points is what lets the platform run efficiently rather than as two disconnected systems.

Intersection point	What the supply side provides	What the demand side consumes
Stock availability	Real-time per-warehouse, per-lot inventory.	Cart validation rejects out-of-stock items at add-to-cart, not at checkout.
Pricing & tiers	Vendor-set retail and wholesale tier prices, plus contract overrides.	Cart applies the right tier automatically based on quantity and buyer type.
Lead time	Production / dispatch / delivery time per warehouse and per category.	Customer sees "arrives by Thursday" before paying, not after.
Geographic routing	Stock per warehouse with GPS coordinates.	Order routes to nearest stocked warehouse; delivery fee calculated from there.
Quality signals	Vendor verification tier, review history, dispute rate.	Search ranking and trust badges visible on every listing card.
Restock	Reorder alerts, sales-velocity data.	Discovery surfaces "back in stock" notifications to customers who'd searched while out.
B2B vs retail	Wholesale pool quantity carved out of total stock.	B2B-mode customer sees deeper inventory; retail customer sees retail-pool only.

Three of these intersections deserve special attention because they're where most marketplaces in emerging markets quietly fail:

- Stock-validation timing. Validating only at checkout is too late — customers who got far enough to enter payment details and then see "out of stock" rarely come back. Validate at add-to-cart, re-validate at checkout, and reserve stock during the checkout window (10–15 minutes per the Day 23-24 spec).
- Lead-time honesty. Promising 2-day delivery and taking 5 destroys reputation faster than anything else. The platform should always quote slightly conservatively. "Arrives Tuesday-Thursday" beats "Arrives Tuesday" if you can only hit Tuesday 60% of the time.
- Geographic routing. A customer in Ndola buying from a vendor with stock in both Lusaka and Ndola should automatically be served from the Ndola warehouse. The customer doesn't choose this; the system does. Without this, multi-warehouse vendors look slower than single-warehouse competitors and the inventory advantage evaporates.

9. Phasing the B2B and Inventory Layers

Your 60-day roadmap nails the B2C foundation. This section maps when the B2B and advanced-inventory features layer on top, without disrupting the launch.

Phase 1 — Day 60 launch (B2C complete, B2B latent in schema)

B2B mode is hidden in Phase 1. The data model accommodates it (price_tiers array, business_verified flag on user, wholesale_pool_quantity field on listings) but no UI surfaces it. Single-warehouse inventory only. This is enough for 75-100 vendors and 50-100 orders per month. Don't build what you don't yet need.

- Active: B2C frontend, single-warehouse inventory, retail pricing, basic search-and-attach, escrow, mobile money, three frontends but Vendor in basic mode.
- Latent (in schema, not in UI): tiered pricing arrays, wholesale pool, multi-warehouse, B2B mode toggle.

Phase 2 — Months 3–6 post-launch (B2B mode goes live)

Once retail volume is steady (300-500 orders/month), enable the B2B layer:

- Tiered pricing UI for vendors. They can now set wholesale tiers.
- Business-mode toggle in customer profile, gated behind PACRA verification.
- RFQ flow extended from services-only to bulk-product purchasing.
- Multi-warehouse inventory for Tier 2/3 vendors with multiple branches.
- Lot/batch tracking for FMCG and pharma vendors.
- Manual tax-compliant invoice generation.

Phase 3 — Months 6–12 (the financial layer)

This is where Convergeo earns the right to expand toward the Alipay-shaped vision:

- Net-30 / Net-60 payment terms for credit-approved B2B buyers. Small-scale lending product. Risk-managed via low initial credit lines and tier requirements.
- Account-manager assignment for top 50 vendors and top 50 B2B buyers — done by you and one operations hire, automated through n8n.
- Contract pricing — pre-negotiated buyer-vendor rates that auto-apply at cart.
- Convergeo Wallet — held balance for vendor payouts that they can either withdraw to mobile money or use to pay platform suppliers (the start of the Alipay-shaped financial loop).
- Demand-forecasting assist with seasonality — replaces the crude reorder-point alerts.

Phase 4 — Year 2 onward (the super-app convergence)

- Full ERP-grade API integration for Tier 3 enterprise vendors.
- Logistics layer maturing into Convergeo Express — own delivery network in major cities, partnered nationwide. Tracks every Convergeo parcel end-to-end.
- Platform credit facility (working capital advances against vendor sales history). This is what Alibaba's Ant Group started as for Taobao sellers.
- Cross-border — Zimbabwe expansion (Q73), eventually full SADC. The B2B layer is what makes cross-border work; consumers won't ship from Zambia to Zimbabwe at scale, but a Zambian wholesaler selling to a Zimbabwean retailer absolutely will.
- AI mode at full strength — semantic discovery, auto-generated product descriptions, pricing optimisation, fraud detection (Q67's prioritised AI features all matured).

10. The Alipay-Shaped Financial Layer (And Why It Comes Last, Not First)

Your prompt explicitly invokes Alipay — and rightly. Alipay is what made Alibaba's empire possible. But Alipay didn't launch as a financial OS. It launched as escrow for Taobao. The financial-OS features grew over a decade, in this order: escrow → wallet → P2P transfers → bill payments → merchant payments → wealth management → lending → insurance → identity → public services.

Convergeo's path follows the same gravity:

Stage 1 — Escrow (already in your plan, Q22)

This is the foundation. The four-step trust visualisation from your wireframes is the Convergeo equivalent of Alipay's original 2003 escrow. Get this right and the rest is downstream.

Stage 2 — Wallet (Year 1)

Once escrow is reliable, vendors and B2B buyers begin holding balances in Convergeo. Withdrawals go to mobile money. Internal transfers between Convergeo accounts are free and instant. This is the Alipay-2005 moment. The wallet is the product, not the payment processor.

Stage 3 — Working capital lending (Year 2)

Once vendors have 6+ months of sales history on the platform, working-capital advances can be offered against future receivables. "You sold K12,000/month average over the last 6 months — borrow up to K8,000 against next month's sales for a 4% fee." Risk-managed because the platform controls the repayment source. This is what Ant Group started as.

Stage 4 — Payment beyond Convergeo (Year 2-3)

Convergeo wallet expanding to QR-pay at non-Convergeo merchants. Bill payments (ZESCO, water, DSTV). Airtime top-ups. P2P transfers. This is the moment Convergeo becomes financial infrastructure, not just a marketplace's payment rail.

Stage 5 — Wealth, insurance, credit scoring (Year 3+)

Group savings products, micro-insurance for vendors, on-platform credit scores that users can carry to other lenders. This is far enough out that it's not worth specifying in detail now. Mention it because it's the destination, but don't engineer toward it on day one.

> **Why this ordering matters:** Every emerging-market platform that tried to launch with the financial layer first failed. The financial layer needs trust capital that only accumulates through millions of successful transactions. Convergeo's 60-day launch gets stage 1 (escrow). Year 1 gets stage 2 (wallet). Year 2 unlocks stage 3 (lending). Anything sooner is borrowing future credibility against present uncertainty, and that's how new fintechs die.

11. Comprehensive Business / Vendor Catalogue for Zambia

This table maps the kinds of business Convergeo's data model and onboarding flows must accommodate. Each row identifies the dominant archetype (from Section 2), the supply mix the business typically offers (products / services / events / inventory), and notes specific to Zambia.

Business type	Archetype	Supply mix	Zambian context
Food & beverage manufacturers	Manufacturer	Products + bulk inventory + B2B	Zambeef, Trade Kings, Parmalat, Zammilk. High-volume FMCG. Multi-warehouse. Wholesale to retailers.
Beverage bottlers	Manufacturer	Products + bulk inventory + B2B	Coca-Cola Beverages Zambia, Heinrich's. Returnable-bottle complexity. Distributor-network heavy.
Maize milling & meal	Manufacturer	Products + bulk inventory + B2B	Sun, Olympic, Antelope. Subsidy-aware pricing. FISP-adjacent.
Cooking-oil producers	Manufacturer	Products + bulk inventory	Cargill, Sun, Sasko presence. Imported + locally pressed.
Brewers	Manufacturer	Products + B2B + events	Zambian Breweries (SAB), local craft. Liquor licensing required.
Cement & building materials	Manufacturer	Products + bulk inventory + B2B + RFQ	Lafarge, Mpande, Sino. Heavy-logistics. RFQ-driven for projects.
Steel & metal fabricators	Manufacturer	Products + made-to-order + B2B	Local welders to industrial-scale rolling mills.
Furniture makers	Manufacturer	Products + made-to-order + B2B	From workshop carpenters (Class E) to factory-scale (Class A/B).
Textile mills & garment factories	Manufacturer	Products + B2B	Mulungushi Textiles. Chitenge fabric production. Export potential.
Soap, detergent, cosmetics manufacturers	Manufacturer	Products + bulk inventory + B2B	Trade Kings, local artisanal brands. Counterfeit-watch zone for Tier 2 only.
Pharmaceutical manufacturers	Manufacturer	Products + B2B (regulated)	Joint ventures (Akums + GRZ 2025 announced). ZAMRA-licensed.

Business type	Archetype	Supply mix	Zambian context
Solar / renewable equipment assemblers	Manufacturer	Products + services + B2B	Booming. Solar kit + installation as bundle.
Fish farms & aquaculture	Producer	Products + B2B (cold chain)	Tilapia, catfish. Lake Kariba & Tanganyika supply. Cold-chain critical.
Poultry farms	Producer	Products + B2B	Day-old chicks, broilers, eggs. High-volume, price-sensitive.
Dairies	Producer	Products + B2B	Parmalat, Zammilk, plus smallholder cooperatives. Cold chain.
Vehicle importers (Japan-route)	Importer	Products (Class D unique) + RFQ	Used-car dealers concentrated in Lusaka. VIN/mileage evidence per Products brief.
Electronics importers	Importer	Products + B2B + bulk inventory	Phones, laptops, appliances. FX-pegged pricing critical.
Pharma importers / wholesalers	Wholesaler	Products (regulated) + B2B	Distribute imported pharma to pharmacies. ZAMRA registration.
FMCG distributors	Wholesaler	Products + bulk inventory + B2B	Multi-brand distributors serving mini-marts and retailers.
Building-materials wholesalers	Wholesaler	Products + bulk inventory + B2B + RFQ	Cement, iron sheets, paints. Project-driven RFQs.
Agro-input distributors	Wholesaler	Products + bulk inventory + B2B	Seeds, fertiliser, pesticides. Strong seasonality. FISP-aware.
Industrial supply wholesalers	Wholesaler	Products + B2B + RFQ	Mining-adjacent supplies, factory consumables. Copperbelt-heavy.
Office-supply wholesalers	Wholesaler	Products + B2B + bulk inventory	Schools, government, corporates. Tender-friendly.
Supermarket chains	Retailer	Products + bulk inventory + occasional events	Shoprite, Pick n Pay, Spar, Choppies. Multi-branch. Bulk SKU catalogue.
Independent mini-marts & groceries	Retailer	Products + walk-in	Township-level. Tier 1/2. Single-shop inventory.
Pharmacies (chain)	Retailer	Products (regulated) + services	Link, Pharmanova. ZAMRA-licensed.

Business type	Archetype	Supply mix	Zambian context
Pharmacies (independent)	Retailer	Products (regulated) + services	Local. Same regulation.
Hardware retailers	Retailer	Products + B2B + RFQ	Builders Warehouse-style. Project & contractor sales.
Electronics retailers	Retailer	Products + services (warranty/repair)	Game Stores, OK Furniture, plus independents.
Fashion retailers (formal)	Retailer	Products	Mr Price, Pep, Edgars-style chains.
Furniture retailers	Retailer	Products + made-to-order + delivery service	Imported + locally made mix.
Auto-parts retailers	Retailer	Products + services (fitment)	Industrial Area concentration.
Salaula traders	Market Trader	Products (Class D unique)	Massive informal category. Each item unique-listing.
Fresh-produce market vendors	Market Trader	Products (Class C commodity)	Soweto, Chisokone. Per-bunch / per-heap pricing. Same-day fulfilment.
Butchery operators	Market Trader / Retailer	Products (Class C, per-kg)	Independent + supermarket counters. Cold-chain.
Phone-accessory stalls	Market Trader	Products + repair services	Cases, chargers, screen protectors. Counterfeit-watch.
Charcoal & firewood vendors	Market Trader	Products (Class C, per-bag)	Hyper-local. Transport-cost dominated.
Salons & barbers	Service Professional	Services (bookable) + retail products	Hair products often retailed alongside service. See Services brief.
Restaurants & cafes	Service / Retail hybrid	Services (walk-in) + delivery + events	Hybrid. Menu = product catalogue. Delivery via platform/partner.
Lodges & guesthouses	Service Professional	Services (reservation) + events (sometimes)	Tourism-tied. Multi-day pricing. Livingstone-heavy.
Tour operators	Service Professional	Services + events (tours-as-events) + ticketing	Adventure activities, safaris. Reservation flow.
Construction contractors	Service Professional	Services + RFQ + B2B materials buyer	Quote-driven. NCC-registered. Sources materials via Convergeo B2B.

Business type	Archetype	Supply mix	Zambian context
Mechanics & auto-service	Service Professional	Services + parts retail + RFQ	Bookable services + parts catalogue.
Lawyers, accountants, consultants	Service Professional	Services (bookable / retainer) + RFQ	Tier 2 mandatory. LAZ/ZICA-regulated.
Photographers & creatives	Service Professional	Services + made-to-order products	Wedding photography, branded merch.
Logistics & courier services	Service / Manufacturer hybrid	Services + B2B partnerships	Last-mile + inter-city. Yango-style + local.
Educational institutions	Service Professional	Services (subscription) + events (open days)	Schools, colleges. Discovery + registration.
Tutoring & training providers	Service Professional	Services + events (workshops)	Bookable individual + group event.
Concert promoters	Event Organiser	Events (ticketed) + ancillary services	Tier 2/3. Pre-event verification for high-value events.
Conference / expo organisers	Event Organiser	Events (multi-day ticketed) + B2B sponsorship	Annual flagships + monthly business events.
Wedding planners & venues	Service / Event hybrid	Services (planning) + events (private) + RFQ	High-value, sensitive UX. Vendor coordination.
Workshop facilitators	Event Organiser / Service Pro	Events (recurring) + services (1-on-1)	Phase 1 launch category per Events brief.
Religious organisations	Event Organiser	Events (free RSVP / ticketed)	Major Zambian category. Free RSVP at scale.
Real estate agencies	Service Professional	Services + listing-based products	ZIEA-registered. Properties as bookmarkable listings.
Insurance brokers	Service Professional	Services + quote	PIA-regulated. RFQ-friendly.
Microfinance & lenders	Financial Services	Services (credit application via Convergeo as discovery only initially)	BoZ-regulated. Convergeo lists, lenders process off-platform until Year 2+.
NGOs & development orgs	Mixed	Services + events + RFQ for procurement	Tender / RFQ heavy. Donor-funded.

Business type	Archetype	Supply mix	Zambian context
Government suppliers	B2B specialised	Products + services + RFQ + tender	Public procurement. Compliant invoicing required. Phase 3+.

That's roughly 60 business types across all six archetypes. Combined with the catalogues from the Services, Products, and Events briefs, Convergeo's full vendor universe is now mapped: 60+ business types, 16 behavioural patterns (5 product classes + 6 service archetypes + 5 event types), 275+ supply sub-categories, three frontends, two pipelines, one schema.

12. Pulling It Together

This is the brief that closes the loop on the Convergeo strategy stack. Services, Products, Events and now Businesses + Pipelines + Frontends — taken together, they describe a platform whose architectural completeness genuinely justifies the Alibaba-Alipay benchmark, scaled to Zambia and to a $2K bootstrap budget.

The seven things to take away:

- The Alibaba four-platform model (1688 / Taobao / Tmall / Alibaba.com) becomes Convergeo's three-frontend model (Customer / Vendor / Admin) plus a B2B mode toggle that lets one platform serve consumer retail, premium retail, B2B wholesale, and procurement-style RFQ — all from the same database.
- Six business archetypes cover the vendor universe: Market Trader, Registered Retailer, Service Professional, Manufacturer, Importer/Wholesaler, Event Organiser. Onboarding answers "what kind of business are you?" once and the dashboard configures itself accordingly.
- Convergeo supports each business across products, services, events, and inventory — with a warehouse-aware inventory model (multi-warehouse, lot/batch, wholesale-pool, FX-pegged pricing, production lead times) for vendors who hold stock at scale.
- Three frontends, each purpose-built. Customer PWA = consumer-facing, fast, low-friction. Vendor app = dense, dual mobile/desktop, business workbench. Admin panel = operational nerve centre that makes the solo-founder + AI tools (Q63) viable.
- The supply pipeline (9 stages, production to listing-ready) and the demand pipeline (12 stages, intent to in-hand) intersect at seven specific points — stock availability, pricing tiers, lead time, geographic routing, quality signals, restock signals, B2B/retail split. Getting these intersections right is what makes the platform feel coherent rather than disjointed.
- B2B mechanics — tiered pricing, MOQs, RFQs, Net-30 terms, account managers, contract pricing — live in the schema from day one but only surface in UI in Phase 2 (months 3-6 post-launch). Phase 3 (months 6-12) layers in the financial features that begin the Alipay-shaped expansion.
- The Alipay parallel is real but the timeline is honest. Escrow now (Q22, day one). Wallet in Year 1. Working-capital lending in Year 2. Off-platform payments in Year 2-3. Wealth and insurance from Year 3+. Every fintech that tried to compress this got buried by trust deficits; Convergeo's 60-day MVP only needs Stage 1, and that's enough.

Across all four briefs the architecture now describes:

- 60+ business types served
- 16 behavioural patterns: 5 product classes, 6 service archetypes, 5 event types
- 275+ supply sub-categories across services, products, and events
- Three frontends sharing one Django REST API

- Two pipelines (supply and demand) meeting at seven explicit intersections
- One schema, one search, one trust system, one payment rail
- Built and shipped in 60 days on a $2K budget by a solo founder using AI tools

That last bullet is the actual punchline. The Alibaba/Alipay vision in your prompt isn't unreasonable for Convergeo — what's unreasonable is doing it the way Alibaba did, with thousands of engineers and twenty years. The way to reach the same destination in Zambia is the architecture in this stack of briefs: simpler than four platforms, smarter than one, and engineered for the constraints that actually exist on the ground here.

End of brief. The strategy stack is complete.

CONVERGEO

Multi-Vendor Commerce & Services Platform for Zambia

Architecture, Segmentation Strategy & Service Catalogue

Strategy & Technical Brief

Prepared: April 2026
Scope: Lusaka, Copperbelt, and nationwide Zambia rollout

1. Executive Framing

Convergeo is positioned as a national hub that unifies products, services, events, and inventory under one discoverable surface. The fundamental challenge is heterogeneity: a salon, a clinic, a butchery and a printing shop each have different inventory models, different booking patterns, different pricing rules, and different trust requirements from a customer. A flat e-commerce schema will fail them all.

The platform must therefore do three things at once: give every vendor a workspace that feels purpose-built for their trade, give every customer a single search experience that resolves intent ("gym near me", "lodge in Livingstone with parking", "who can print 200 banners by Friday"), and give the operator (you) one underlying data model that scales across categories and towns without ballooning into hundreds of bespoke modules.

This document lays out how to do that. It covers the segmentation model, the vendor experience, the customer discovery layer including vector + geo search, payment realities specific to Zambia, and a comprehensive catalogue of the service categories Convergeo should support at launch and over time.

Why this matters now

Three signals make Zambia a strong fit for this kind of platform in 2026:

- Mobile money penetration is high. MTN Mobile Money, Airtel Money and Zamtel Kwacha are the de facto rails for retail payments — well over half of adult Zambians transact through them, and monthly volumes have reached the hundreds of billions of kwacha. Any platform that doesn't natively accept these will not convert.
- Services contribute roughly 58% of GDP, while informality covers around 70% of employment. There is an enormous tail of unlisted, unsearchable small businesses — exactly the inventory a hub aggregator monetises.
- Existing online directories in Zambia are static yellow-pages-style listings. There is no incumbent that combines vendor self-service, real-time bookings, geo-search, and integrated mobile money checkout.

2. Core Architectural Decisions

Before getting into categories and search, four architectural decisions shape everything else. Get these right early — they are expensive to undo later.

2.1 The "Listing" is the universal primitive

Don't model Products, Services, Events and Inventory as four separate top-level concepts. Model one polymorphic Listing entity with a discriminator field (kind: product | service | event | rental | space | menu_item) and let category-specific attributes live in a flexible attributes JSON column. This gives you:

- One search index, one cart, one review system, one analytics pipeline.
- The ability to add a new sub-type (e.g. "course enrollment", "tour package") without schema migrations.
- A clean place to put per-category fields: a salon service has duration_minutes and staff_required; a lodge listing has check_in_time, max_guests, amenities[]; a butchery product has cut_type, weight_grams, price_per_kg.

Underneath, the relational core stays normalised: vendors, locations, bookings, orders, payments, reviews, and media. Only the variable category attributes live in JSON.

2.2 Vendor → Location → Listing, not Vendor → Listing

Many Zambian businesses run multiple branches: a salon chain in Lusaka and Kitwe, a restaurant with three outlets, a clinic with a head office and two satellites. Putting Location as a first-class entity between Vendor and Listing solves a lot of pain:

- Each branch has its own opening hours, staff roster, phone number, GPS coordinates, and bookable inventory.
- A customer searching "physiotherapist in Kabwata" gets the Kabwata branch ranked above the Northmead one even though both belong to the same vendor.
- Vendor-level branding (logo, colour theme, about page) inherits down, but location-level overrides are allowed.

2.3 Two-layer taxonomy: Category + Tags

A rigid single hierarchy (Health → Clinics → Dental → Orthodontics) breaks the moment a vendor doesn't fit cleanly. Use a hybrid:

- A controlled Category tree, two to three levels deep, used for navigation, filters, and homepage browsing. This is curated by you and changes rarely.

- Free-form Tags applied by the vendor and moderated, used for search recall. A gym tags itself "crossfit", "women only", "24/7", "sauna" — none of which need to be official categories.

Search ranking blends both: category match is a strong signal, tag match is a softer signal, and the embedding-based semantic match (next section) catches everything in between.

2.4 Vendor isolation: their dashboard is their business

Every vendor must feel like Convergeo is theirs, not a flea-market stall. The dashboard exposes:

- Branding controls — logo, cover photo, theme accent colour, about copy, social links.
- Catalogue management — add/edit listings, bulk price updates, stock levels for products, service-duration and buffer-time for services.
- Bookings & availability — calendar view, staff assignment, blackout dates, recurring availability rules.
- Orders & payouts — order queue, fulfilment status, mobile-money settlement reports, downloadable invoices.
- Insights — page views, search impressions, conversion funnel, top-searched terms that landed on their page (this is the data they can't get anywhere else and it's why they'll stay).
- Communication — in-platform chat with customers, SMS/WhatsApp templated responses, automated booking reminders.

3. Segregating Many Different Services Cleanly

The trap most multi-vendor platforms fall into is forcing every business through the same checkout flow. A haircut, a hospital consultation, a printing job, a kilo of beef and a weekend at a lodge are not the same transaction. Segregation needs to happen at three layers.

3.1 Layer 1: Transaction archetypes

Every listing on Convergeo collapses into one of six transaction archetypes. The archetype determines the checkout flow, the vendor's tooling, and the customer's expectations:

Archetype	Description	Examples
Buy-now product	Physical or digital good, instant purchase, fulfilment via delivery or pickup.	Butchery cuts, supermarket items, leather goods, printed merchandise, electronics.
Bookable service	Time-slotted appointment with a provider; capacity is the provider's calendar.	Salon, dental, physio, mechanic, tutoring, photography session.
Reservation / stay	Date-range booking against limited inventory units.	Lodges, guesthouses, conference rooms, equipment rental, car hire.
Walk-in / on-demand	Customer arrives without a slot; platform handles discovery only, payment is on-site or via QR.	Restaurants (dine-in), bars, gyms (day pass), pharmacies, fuel stations.
Quote-based	Job is too variable to price up front; vendor receives request and replies with a quote.	Construction, large catering, custom printing runs, bulk leather work, event planning.
Event ticket	Time-fixed, capacity-limited admission.	Concerts, conferences, weddings as venues, sports fixtures, workshops.

The vendor onboarding wizard asks one question first — "what kind of business are you?" — and from the answer derives which archetypes are available, which fields are mandatory, and what their dashboard defaults look like. A salon owner never sees a SKU field; a butcher never sees a calendar.

3.2 Layer 2: Industry verticals (the visible category tree)

On top of archetypes sits the customer-facing vertical taxonomy. Section 5 of this report contains the comprehensive catalogue, but the top level is intentionally short — about a dozen verticals — so that the homepage and mobile navigation stay scannable. Going deeper than three levels in a tree starts to hurt: customers don't browse, they search. The tree exists for SEO landing pages and filtering, not for navigation as a primary discovery method.

3.3 Layer 3: Vendor-controlled storefront

Within their storefront, the vendor controls how their offerings are grouped. A restaurant might use sections ("Starters", "Mains", "Drinks"); a salon uses service categories ("Hair", "Nails", "Spa"); a printing shop might use job types ("Business cards", "Banners", "T-shirts"). The platform provides flexible grouping primitives — Sections, Collections, Menus — and the vendor picks which one fits.

3.4 Trust and verification tiers

Not every category carries the same regulatory weight. A clinic offering medical services cannot be onboarded with the same self-service flow as a phone-case seller. Bake verification into the listing surface itself:

- Tier 0 — Self-listed: instant onboarding, marked clearly as "Unverified vendor". Suitable for low-risk products and informal services.
- Tier 1 — Identity verified: NRC and PACRA business registration confirmed. Visible badge.
- Tier 2 — Sector verified: regulator licence on file (HPCZ for clinics, ZMRA for transport, ZAMRA for pharmacies, Tourism Council for lodges). Required, not optional, for those categories.
- Tier 3 — Convergeo Preferred: consistent ratings, completed orders threshold, response-time SLA met. Earns ranking boost and a visible badge.

Showing the badge prominently next to the vendor's name is one of the highest-leverage trust mechanics you can deploy in a market where buyers are reasonably wary of online scams.

4. Discovery Layer: Vectorised RAG + Proximity Search

This is the part you specifically asked to deep-dive on. The objective: a stranger arrives in Solwezi, opens Convergeo, types "gym" or "swimming pool" or "somewhere quiet to work with wifi", and gets the right answer ranked by relevance and distance, even when the vendor never used those exact words.

4.1 Why pure keyword search isn't enough

Keyword search fails on three patterns that are extremely common in Zambia:

- Synonym drift. A customer searches "barber" but vendors describe themselves as "hair stylist", "men's grooming", or "kinyozi". Keyword indexes miss this.
- Local language and code-switching. "Nshima place", "chibuku bar", "chitenge tailor" — these terms barely appear in formal listings but are how people actually search.
- Intent rather than name. "Place to take my kids on Saturday" or "somewhere to work with wifi" describes a need, not a category.

Vector embeddings handle all three because they place semantically similar phrases close together in a high-dimensional space, regardless of exact wording.

4.2 Recommended discovery pipeline

A practical pipeline for Convergeo, in order:

Step 1 — Ingestion: build the listing's semantic document

For every listing, concatenate a structured "semantic document" of: vendor name, category path, tags, description, top reviews, location name, neighbourhood, town, and the vendor's own free-text answer to "in one paragraph, what would a customer tell a friend about you?". This last field is gold — vendors describe themselves in customer language.

Generate an embedding from this document using a multilingual model (OpenAI text-embedding-3-large, Voyage AI, or Cohere embed-multilingual — all handle English well and cope with code-switched local terms reasonably). Store the vector in pgvector if you're on Postgres, or in a dedicated store like Qdrant or Weaviate if you anticipate hundreds of thousands of listings.

Step 2 — Query expansion

When a user types "gym", run the query through a lightweight LLM call (or a cached lookup table for common terms) that expands it to: gym, fitness centre, workout, weights, crossfit, aerobics, personal trainer. The expanded query gets embedded and searched. Cache aggressively — most search terms repeat.

Step 3 — Hybrid retrieval

Run two searches in parallel and fuse the results:

- Vector search over the embedding store, returning the top 50 listings by semantic similarity, filtered to a generous geo-radius (say 50km from the user) and to active vendors only.
- Lexical search (BM25 / Postgres full-text / Meilisearch) over name, tags and category — this catches exact-match cases where someone searches a vendor's actual name.

Reciprocal rank fusion (RRF) is the simplest and most robust way to merge the two ranked lists. Don't overthink this — RRF beats most learned rankers in low-data regimes.

Step 4 — Geo and context re-ranking

Take the fused top 50 and re-rank using a weighted score:

- **Distance score:** inversely proportional to road distance from the user. Use the Haversine formula for the cheap version, or a routing API (OSRM self-hosted, or Google Distance Matrix) for the accurate one. Cap the bonus at ~10km — beyond that, distance matters less than relevance.
- **Open-now score:** a listing whose hours say it's open right now gets a multiplicative boost. "Closed for the night" listings drop down but stay visible.
- **Quality score:** rating × log(review count). Use the Bayesian average to avoid five-star vendors with only one review out-ranking 4.7-star vendors with two hundred.
- **Verification score:** Tier 2 and Tier 3 vendors get a small boost, especially in regulated categories (medical, financial).
- **Personalisation:** if the user has a history, modest weight on previously-engaged categories. Be conservative — over-personalisation in a discovery context is annoying.

Step 5 — Presentation: map + list together

The results screen should show a list and a map side by side (desktop) or stacked with a toggle (mobile). Each result card carries: photo, name, distance, rating, opening status, a 2-line snippet describing what they offer, and price range. Clicking pins on the map filters the list and vice-versa.

4.3 Proximity-first user flows

Three flows that take advantage of the geo layer and that are particularly valuable in a country where towns are spread out:

- "New in town" mode: detect when a user's location is more than 100km from their previous session location. Switch the homepage to a "Welcome to Solwezi" layout featuring essentials — supermarkets, fuel, pharmacies, lodges, restaurants, ATMs — instead of the usual personalised feed.

- "Around me right now": a one-tap button that drops keyword search entirely and shows everything within 2km, grouped by category. Useful when you don't know what you're looking for.
- "On the route": for inter-city travellers, given an origin and destination, surface lodges, fuel stations, restaurants, and tyre repair shops along the route corridor. This is genuinely differentiated — Google Maps does this for global brands but not for the long tail of Zambian SMEs.

4.4 Conversational front-end (Convergeo AI assistant)

Once the RAG infrastructure is in place, exposing it through a chat interface is almost free. The user types or voice-inputs "I need a printing shop that can do 500 wedding invitations by Saturday in Ndola" and the assistant:

- Parses the entities (item, quantity, deadline, location).
- Runs the discovery pipeline above with appropriate filters (category=printing, town=Ndola).
- Pings the top three vendors via the platform's quote-request flow.
- Returns the matches in chat, with one-tap actions to message, call, or request a quote.

Voice input matters more than it does in higher-income markets — typing on cheap Android keyboards is slower than speaking, and voice is a strong accessibility win for users with limited literacy.

5. Comprehensive Service & Vendor Catalogue for Zambia

The table below is a working catalogue of the vendor categories Convergeo should be designed to absorb. It's organised by vertical, and for each sub-category it lists the dominant transaction archetype (from Section 3.1), illustrative Zambian examples or contexts, and the regulatory body whose licence will be required for Tier 2 verification where relevant.

Read this not as a launch list — that would be too much — but as the universe the platform must architect for. Section 6 covers phasing.

Vertical	Sub-category	Archetype	Zambian context	Regulator
Beauty & Personal Care	Hair salons & barbers	Bookable	Independent kinyozis, chain salons in malls (Levy, Manda Hill), township braiding studios.	—
Beauty & Personal Care	Nail technicians & spas	Bookable	Lusaka and Copperbelt urban; growing in Solwezi and Kitwe.	—
Beauty & Personal Care	Make-up artists	Bookable / Quote	Wedding-heavy demand, freelance MUAs across cities.	—
Beauty & Personal Care	Massage & wellness therapists	Bookable	Hotel-based and standalone wellness studios.	HPCZ if therapeutic
Health & Medical	General practitioners & clinics	Bookable	Private clinics in Lusaka, mission hospitals nationwide.	HPCZ
Health & Medical	Dental practices	Bookable	Limited supply outside major towns — strong discovery use case.	HPCZ
Health & Medical	Optometrists & opticians	Bookable / Buy-now	Eye-care chains plus standalone practices.	HPCZ
Health & Medical	Physiotherapy & chiropractic	Bookable	Sports and post-injury demand.	HPCZ
Health & Medical	Diagnostic & imaging labs	Bookable	Pathology, X-ray, ultrasound; often referred.	HPCZ / ZAMRA

Vertical	Sub-category	Archetype	Zambian context	Regulator
Health & Medical	Pharmacies	Buy-now / Walk-in	Independent and chain (Link, Pharmanova). Prescription handling required.	ZAMRA
Health & Medical	Mental health & counselling	Bookable	Underserved nationally; teletherapy a fit.	HPCZ
Health & Medical	Veterinary clinics	Bookable	Urban pet care, also livestock-focused in farming districts.	Vet Council of Zambia
Food & Hospitality	Restaurants & cafes	Walk-in / Buy-now	From street-food nshima joints to fine dining; delivery via in-platform or partner.	Local council, ZABS
Food & Hospitality	Bars & lounges	Walk-in	Lusaka nightlife corridors, township pubs.	Liquor Licensing Board
Food & Hospitality	Catering & event food	Quote	Weddings, corporate events, kitchen parties.	Local council
Food & Hospitality	Lodges & guesthouses	Reservation	Heavy along Livingstone, South Luangwa, Lower Zambezi corridors.	Tourism Council of Zambia
Food & Hospitality	Hotels	Reservation	Branded chains in Lusaka/Copperbelt, independents elsewhere.	Tourism Council of Zambia
Food & Hospitality	Self-catering apartments	Reservation	Growing alternative to hotels in Lusaka.	Tourism Council of Zambia
Food & Hospitality	Conference & event venues	Reservation / Quote	Hotel ballrooms, dedicated conference centres, garden venues.	—
Food & Hospitality	Bakeries & confectioneries	Buy-now / Quote	Cake-makers, custom orders for celebrations.	Local council
Food Retail & Production	Butcheries	Buy-now	Independent neighbourhood butcheries plus supermarket counters.	Ministry of Health, ZABS
Food Retail & Production	Fishmongers & fish farms	Buy-now / Quote	Lake Kariba and Lake Tanganyika supply chains.	DoF

Vertical	Sub-category	Archetype	Zambian context	Regulator
Food Retail & Production	Dairies	Buy-now	Local milk and yoghurt producers.	ZABS
Food Retail & Production	Greengrocers & farm-to-table	Buy-now	Soweto Market vendors; emerging direct-from-farm subscriptions.	—
Food Retail & Production	Supermarkets & mini-marts	Buy-now / Walk-in	Shoprite, Pick n Pay, Spar plus thousands of local groceries.	—
Food Retail & Production	Salaula & fresh produce markets	Walk-in	City Market, Soweto, Chisokone — discovery and stall-level listings.	Local council
Fitness & Sport	Gyms & fitness centres	Walk-in / Subscription	Day-pass and membership models. Lusaka, Kitwe, Ndola dense.	—
Fitness & Sport	Swimming pools	Walk-in	Hotel pools open to public, sports clubs, schools after hours.	—
Fitness & Sport	Yoga & pilates studios	Bookable / Subscription	Lusaka emerging market.	—
Fitness & Sport	Sports clubs (golf, tennis, squash)	Reservation / Subscription	Lusaka Golf Club, Ndola Country Club.	—
Fitness & Sport	Personal trainers	Bookable	Often gym-affiliated, increasingly independent.	—
Fitness & Sport	Sports academies & coaching	Subscription	Football, netball, swimming youth academies.	—
Automotive & Transport	Mechanics & service centres	Bookable / Quote	Heavy in Light Industrial Area, Kafue Road.	RTSA
Automotive & Transport	Tyre & battery shops	Buy-now / Walk-in	Highway corridors particularly important.	—
Automotive & Transport	Car wash & detailing	Walk-in / Bookable	Often informal — formalisation opportunity.	—
Automotive & Transport	Auto parts retailers	Buy-now	Genuine and aftermarket; bulk demand in mining towns.	—

Vertical	Sub-category	Archetype	Zambian context	Regulator
Automotive & Transport	Car dealerships (new & used)	Quote	Japan-import dealers concentrated in Lusaka.	RTSA
Automotive & Transport	Car hire	Reservation	Airport-led market; emerging peer-to-peer.	RTSA
Automotive & Transport	Driving schools	Bookable / Subscription	RTSA-mandated training providers.	RTSA
Automotive & Transport	Logistics & courier	Quote / Buy-now	Last-mile in cities, inter-city trucking, cross-border.	RTSA
Automotive & Transport	Boda-boda & motorcycle taxis	On-demand	Township and peri-urban backbone.	RTSA
Home Services	Plumbers	Quote / Bookable	Pure on-demand; verified-vendor trust signal critical.	—
Home Services	Electricians	Quote / Bookable	ERB-certified for solar work increasingly common.	ERB for solar
Home Services	Carpenters & joinery	Quote	Custom furniture is a real category — overlaps with manufacturing.	—
Home Services	Painters & decorators	Quote	—	—
Home Services	Cleaning services	Bookable / Subscription	Domestic and commercial; recurring revenue model.	—
Home Services	Pest control	Bookable	—	ZEMA
Home Services	Solar installers	Quote	Booming sector given load-shedding and rural off-grid demand.	ERB
Home Services	Borehole drilling & water	Quote	Peri-urban Lusaka especially.	WARMA
Home Services	Landscaping & gardening	Bookable / Subscription	Residential and commercial.	—
Home Services	Security & CCTV installation	Quote	Rapidly growing residential demand.	PSIRA-type registration

Vertical	Sub-category	Archetype	Zambian context	Regulator
Construction & Real Estate	Building contractors	Quote	NCC-registered grades 1–6 critical to surface in vendor profile.	NCC
Construction & Real Estate	Architects	Quote	ZIA-registered.	ZIA
Construction & Real Estate	Quantity surveyors & engineers	Quote	—	EIZ / SIZ
Construction & Real Estate	Hardware & building supplies	Buy-now	Cement, steel, roofing — high-volume retail.	—
Construction & Real Estate	Real estate agents	Listing-based	Property sales and rentals; Pam Golding, Knight Frank, plus many independents.	ZIEA
Construction & Real Estate	Property management	Subscription / Quote	Landlord-facing service.	ZIEA
Construction & Real Estate	Interior designers	Quote	—	—
Education & Training	Schools (public & private)	Subscription	ECCE through secondary; admissions discovery is the use case.	Ministry of Education
Education & Training	Colleges & universities	Subscription	UNZA, CBU, plus dozens of private institutions.	HEA / TEVETA
Education & Training	Tutoring & exam prep	Bookable	Cambridge, ECZ, professional exam prep — strong online fit.	—
Education & Training	Driving schools	Bookable	Already covered under Automotive — listed cross-vertical.	RTSA
Education & Training	Skills & vocational training	Subscription	TEVETA-accredited centres.	TEVETA
Education & Training	Music & arts lessons	Bookable	—	—
Education & Training	Language schools	Subscription	English, French, Mandarin in Lusaka.	—

Vertical	Sub-category	Archetype	Zambian context	Regulator
Education & Training	Online courses & coaching	Subscription / Buy-now	Local creators, professional certifications.	—
Professional Services	Lawyers & legal services	Bookable / Quote	LAZ-regulated; Tier 2 verification mandatory.	LAZ
Professional Services	Accountants & tax consultants	Bookable / Quote	ZICA-registered.	ZICA
Professional Services	Business registration & PACRA agents	Quote	Common SME need.	PACRA
Professional Services	Insurance brokers	Quote	Pension, motor, health, life.	PIA
Professional Services	Marketing & PR agencies	Quote	—	—
Professional Services	Translation & interpretation	Quote	Local languages and Portuguese for cross-border.	—
Professional Services	Recruitment & staffing	Quote	—	—
Professional Services	Notary & document services	Bookable	Commissioner of Oaths services in high demand.	LAZ
Creative & Media	Photographers & videographers	Bookable / Quote	Wedding, corporate, and fashion shoots.	—
Creative & Media	Graphic designers	Quote	—	—
Creative & Media	Web & app developers	Quote	Local agency scene growing.	—
Creative & Media	Printing shops	Quote / Buy-now	Banners, business cards, branded merch — heavy SME demand.	—
Creative & Media	Sign-writing & branding	Quote	Vehicle wraps, shopfront signage.	—
Creative & Media	DJs & musicians	Bookable	Weddings, kitchen parties, corporate.	—

Vertical	Sub-category	Archetype	Zambian context	Regulator
Creative & Media	MCs & event hosts	Bookable	—	—
Creative & Media	Event planners & decorators	Quote	Major weddings industry.	—
Agriculture	Agro-dealers (seeds, fertiliser)	Buy-now	Critical for smallholder farmers; FISP-adjacent.	ZARI / Ministry of Agriculture
Agriculture	Veterinary supplies	Buy-now	—	Vet Council
Agriculture	Farm equipment & tractors	Quote / Buy-now	Sales, hire, repair.	—
Agriculture	Poultry & livestock suppliers	Buy-now / Quote	Day-old chicks, feed, breeding stock.	—
Agriculture	Irrigation & farm services	Quote	Boreholes, drip, sprinkler — overlaps with home services.	WARMA
Agriculture	Agronomy & extension services	Bookable / Quote	Growing private advisory market.	—
Crafts & Manufacturing	Tailors & dressmakers	Quote / Bookable	Chitenge fashion is a major category.	—
Crafts & Manufacturing	Leather workers & cobblers	Buy-now / Quote	Belts, bags, shoe repair — informal sector.	
Crafts & Manufacturing	Curio & art galleries	Buy-now	Tourist-driven, Lusaka and Livingstone.	—
Crafts & Manufacturing	Furniture makers	Quote / Buy-now	Custom and ready-made.	—
Crafts & Manufacturing	Welders & metal fabricators	Quote	Gates, burglar bars, trailers.	—
Crafts & Manufacturing	Pottery, basketry, beadwork	Buy-now	Cultural artisans; export potential.	—
Financial & Digital	Mobile money agents	Walk-in	Float, deposit, withdrawal points — discovery is the value.	BoZ

Vertical	Sub-category	Archetype	Zambian context	Regulator
Financial & Digital	Microfinance & lending	Quote	—	BoZ
Financial & Digital	Forex bureaux	Walk-in	—	BoZ
Financial & Digital	Insurance providers	Quote	Direct-to-consumer products.	PIA
Financial & Digital	IT support & repair	Bookable / Quote	Phone, laptop, network.	—
Financial & Digital	Internet service providers	Subscription	Liquid, Paratus, Zamtel, MTN, Airtel — installation requests.	ZICTA
Financial & Digital	Cyber cafes & print bureaus	Walk-in	Government-form printing demand significant.	—
Tourism & Leisure	Tour operators	Reservation / Quote	Safaris, falls tours, walking safaris.	Tourism Council of Zambia
Tourism & Leisure	Tour guides (independent)	Bookable	Major opportunity to formalise informal guides.	Tourism Council of Zambia
Tourism & Leisure	Adventure activities	Bookable	Bungee, rafting, helicopter — Livingstone-centric.	Tourism Council of Zambia
Tourism & Leisure	Cinemas & entertainment venues	Buy-now	Ster-Kinekor and emerging local screens.	—
Tourism & Leisure	Game parks & reserves	Reservation	—	DNPW
Tourism & Leisure	Boat hire & water sports	Reservation	Kariba, Zambezi, Lake Tanganyika.	Department of Maritime
Events	Concerts & festivals	Event ticket	Major demand pent-up; Lusaka, Livingstone festivals.	—
Events	Conferences & expos	Event ticket	Business and professional events.	—
Events	Sports events & fixtures	Event ticket	Football, rugby, athletics.	—

Vertical	Sub-category	Archetype	Zambian context	Regulator
Events	Workshops & masterclasses	Event ticket	Professional development.	—
Events	Religious & community events	Event ticket / Free RSVP	Often free but discovery valuable.	—
Public & Misc	Funeral services	Quote	Coffin makers, hearses, burial planning — sensitive but real demand.	—
Public & Misc	Religious institutions	Listing-based	Churches, mosques — discovery only.	—
Public & Misc	Animal services & boarding	Bookable	Kennels, grooming, training.	Vet Council
Public & Misc	Childcare & nannies	Bookable / Quote	—	—
Public & Misc	Elder care	Bookable / Quote	Emerging category as urban families nuclearise.	—
Public & Misc	Errand & concierge services	Bookable	Queue-running for government offices, bills, etc. — strong fit.	—

That's roughly 110 sub-categories across 14 verticals. Not all need to launch on day one (Section 6), but the data model and category tree must accommodate all of them from the start to avoid migrations later.

6. Recommended Launch Phasing

Trying to onboard 110 sub-categories at launch will spread effort thin and produce a half-empty platform that nobody trusts. The categories below are sequenced by a combination of demand intensity, vendor accessibility, and how well-suited they are to the bookable/buy-now flows that are easiest to build.

Phase 1 — Lusaka launch (months 0–4)

Eight high-density, high-demand verticals where you can hand-recruit the first 200–300 vendors and prove the model:

- Beauty & personal care (salons, barbers, nails)
- Restaurants, cafes, bars (walk-in + delivery)
- Lodges and guesthouses
- Gyms and fitness
- Mechanics, car wash, tyre shops
- Printing and creative services
- Health: clinics, dental, pharmacies
- Home services: plumbers, electricians, cleaners

Phase 2 — Lusaka deepening + Copperbelt expansion (months 4–9)

Add categories that benefit from the network effect of an active platform:

- Tutoring, training, professional services (lawyers, accountants)
- Events and event-adjacent services (photographers, planners, MCs, DJs)
- Real estate listings (rentals first, then sales)
- Construction trades and contractors
- Tourism and tour operators (Livingstone push)
- Geographic expansion to Kitwe, Ndola, Livingstone

Phase 3 — Long tail and rural reach (months 9–18)

Once supply density and trust are established, open the platform to:

- Agriculture inputs, agro-dealers, farm services
- Funeral services, religious institutions, community events
- Crafts, tailors, leather, furniture

- Mobile money agents, forex, microfinance directories
- Smaller towns: Solwezi, Chipata, Mongu, Kasama
- Conversational AI assistant launch (RAG layer matures)

7. Payments, Trust & Operational Realities

7.1 Payments

Mobile money is the spine. Build native integrations with MTN MoMo, Airtel Money, and Zamtel Kwacha rather than relying solely on a single aggregator — the per-transaction economics are better and the customer experience is faster (USSD push instead of redirects). For card payments and where direct integration is not yet feasible, layer an aggregator: ZynlePay, DPO Pay, Pesapal and Flutterwave all support the Zambian market. Build the abstraction so any of them is swappable.

Two important design choices on payments:

- Hold funds in escrow for bookings and quote-based services until the customer confirms fulfilment, or for a fixed window after the appointment (typically 24–48 hours). This is the single biggest trust mechanic for a new platform — vendors get reliably paid out, customers get protection. Be transparent with both sides about how it works.
- Settle vendor payouts to mobile money or bank account on a predictable schedule (weekly is the right default). Daily payouts sound vendor-friendly but fragment the operations work and increase per-payout fees.

7.2 Trust mechanics beyond verification

- Reviews with order proof. Only customers who completed an order or booking can review. This eliminates 90% of fake-review attacks.
- Response time visible on every vendor profile. "Usually replies within 2 hours" is a stronger trust signal than star ratings on a brand-new listing.
- Dispute resolution with a real human. Even at low volume, a 24-hour SLA on disputes builds outsized reputation.
- Photo evidence on completion. For services like home repairs, a vendor uploading before/after photos to close the job both protects them and creates marketing collateral.

7.3 Connectivity and offline-tolerance

Data is expensive and sometimes unstable, especially outside the line of rail. Three things that materially help:

- A lean mobile web experience under 200KB initial payload. The platform should be usable on a 3G connection on a budget Android.
- USSD integration for the most common vendor actions (check today's bookings, mark order fulfilled). Vendors without smartphones can still operate.

- WhatsApp as a first-class channel for booking confirmations, reminders, and customer-vendor chat. WhatsApp is universal in Zambia in a way that even SMS isn't.

7.4 Localisation details that compound

- Kwacha as default currency, with USD optionally for tourism listings.
- Open hours that respect local rhythm — many businesses close for lunch or have different Sunday hours; the schema should support multi-segment days.
- Address handling: Zambia doesn't have universal street addresses. Use "plot number + neighbourhood + landmark" with mandatory GPS pin. The pin is the source of truth.
- Phone numbers default to +260 with auto-formatting; allow multiple numbers per location since vendors often have separate Airtel/MTN lines for clients on different networks.

8. Pulling It Together

Convergeo's defensibility, in one line, is this: it's the only place in Zambia where any kind of vendor — formal or informal, single shop or chain — can show up, get found by intent rather than by knowing-the-name, transact on the rails customers already use, and accumulate a reputation that travels with them as they grow.

The architectural commitments that deliver that:

- One polymorphic Listing primitive, six transaction archetypes, two-layer taxonomy.
- Vendor → Location → Listing hierarchy so multi-branch businesses model cleanly.
- Hybrid retrieval (BM25 + vector embeddings) fused with geo and quality re-ranking, exposed both as a search bar and as a conversational assistant.
- Native MTN/Airtel/Zamtel mobile money + escrow + aggregator card fallback.
- Verification tiers visible everywhere they matter, with regulator-tied checks for medical, legal, financial, transport and tourism.
- Phased category rollout — eight verticals at Lusaka launch, expanding by demand and supply density, with the data model already wide enough for all 110+ sub-categories from day one.

Build the spine right and Convergeo grows in two dimensions at once: more towns get covered, and more categories within each town fill in. The vector + geo discovery layer is what makes that growth feel coherent to the user no matter how big it gets.

End of brief.